International Journal of Information Management 89 (2026) 103057

Contents lists available at ScienceDirect

International Journal of Information Management

journal homepage: www.elsevier.com/locate/ijinfomgt

Artificial intelligence for information-driven resilience: Enhancing strategic adaptation and entrepreneurial success

Shaofeng Wang [a,c,1], Hao Zhang [b,*,2]

[a] International Business School, Fuzhou University of International Studies and Trade, Fuzhou 350202, China
[b] School of Economics and Management, Zhejiang Shuren University, Hangzhou 310015, China
[c] School of Management, Zhejiang University, Hangzhou 310058, China

ARTICLE INFO

Keywords:
Artificial intelligence
Information management
Strategic resilience
Digital transformation
Entrepreneurial performance
Knowledge management

ABSTRACT

This study investigates how artificial intelligence (AI) innovation capability—the systematic integration, management, and application of AI-driven information—enhances strategic resilience and entrepreneurial success under conditions of market volatility and resource constraints. We conceptualize information-driven resilience as the organizational capacity to convert AI-enabled information flows into adaptive responses that sustain competitive functioning. Drawing on dynamic capability theory, we examine how early-stage ventures transform AI capabilities into adaptive capacity and competitive advantage. Using a multi-method approach, we collected two-wave survey data from 357 early-stage ventures in China and Europe, complemented by importance-performance map analysis (IPMA), fuzzy-set qualitative comparative analysis (fsQCA), and semi-structured interviews. Our convergent multi-method design provides complementary evidence: PLS-SEM establishes the mediation hypothesis, IPMA identifies actionable managerial priorities, fsQCA reveals equifinal configurational pathways to high performance, and interviews illuminate the underlying mechanisms. Specifically, AI innovation capability is significantly associated with strategic resilience, which is positively related to entrepreneurial success. The mediating effect of strategic resilience is robust and fully accounts for the AI capability–success relationship, indicating that the mechanism through which AI investments generate performance returns is fundamentally organizational rather than technological in nature. The value derived from AI is unlocked by first building organizational capacity for adaptive learning and agile decision-making. The study contributes to information management literature by clarifying how AI-enabled information processing capabilities enhance adaptive responses in uncertain environments, while providing practitioners with an actionable framework for developing information-driven resilience through strategic AI implementation.

1. Introduction

In the evolving landscape of information management, organizations face mounting pressure to strategically leverage artificial intelligence (AI)—the systematic integration, deployment, and application of AI technologies across organizational processes to transform data into actionable intelligence and drive adaptive responses to environmental shifts (Chau & Xu, 2025; Gama & Magistretti, 2025; Nambisan et al., 2017). This strategic AI deployment, encompassing technological integration, managerial orchestration, and innovative application, has elevated AI innovation capability to a strategic imperative for organizational survival and competitive positioning (Ameen et al., 2024; Gupta et al., 2025; Przegalinska et al., 2025; Strauss et al., 2024). The dynamic information ecosystems of China and Europe further intensify these pressures, compelling early-stage ventures to implement advanced information processing technologies that enhance competitive positioning through superior knowledge generation and proactive responsiveness (Ku & Chen, 2024; Lengnick-Hall et al., 2011). The selection of China and Europe as our dual research contexts was deliberate and theoretically motivated. These two regions represent contrasting institutional logics that enable us to test the robustness of the AI innovation capability–strategic resilience–entrepreneurial success model across

* Corresponding author.
E-mail addresses: vipwhsl@hotmail.com (S. Wang), zhanghao08042022@163.com (H. Zhang).
[1] https://orcid.org/0000-0002-0300-2453
[2] https://orcid.org/0009-0009-7997-9410

https://doi.org/10.1016/j.ijinfomgt.2026.103057
Received 16 March 2025; Received in revised form 4 March 2026; Accepted 5 March 2026
Available online 10 March 2026

fundamentally different environments. China operates within a state-led innovation ecosystem characterized by strong governmental support for AI development, rapidly scaling digital infrastructure, and an entrepreneurial culture that prioritizes speed of adoption and application innovation. Europe, by contrast, embodies a mature regulatory landscape shaped by stringent data governance frameworks (notably GDPR), market-driven competitive dynamics, and established norms of stakeholder accountability that may place greater emphasis on AI management precision and contextual resilience. By spanning these contrasting institutional environments, our research design allows us to assess whether the core mediation mechanism we identify operates as a context-transcendent phenomenon or is contingent upon specific institutional configurations—a question of direct relevance to dynamic capability theory's generalizability claims (Cavusgil & Deligonul, 2024; Teece et al., 1997). This institutional heterogeneity thus provides unique theoretical leverage for evaluating the boundary conditions and universal applicability of our proposed information-driven resilience model.

Adopting AI-driven information management solutions enables ventures to optimize knowledge flows, anticipate market trends through intelligent data analysis, and identify emergent opportunities when conventional information processing capabilities are constrained by limited resources and compressed timeframes (Chattopadhyay et al., 2024). We conceptualize this strategic AI capacity as AI innovation capability—a multidimensional information management competence encompassing firms' proficiency in integrating AI solutions into information systems (integration innovation), leveraging AI tools for management optimization (management innovation), and deploying AI technologies to identify novel opportunities and develop innovative offerings (application innovation) (Cillo & Rubera, 2024; Mikalef & Gupta, 2021). This three-dimensional conceptualization captures the strategic nature of AI deployment by recognizing that effective AI management requires not merely technological adoption but rather organizational capabilities for purposeful integration, resource orchestration, and opportunity exploitation.

We ground our investigation in dynamic capability theory (Barreto, 2010; Teece et al., 1997; Teece, 2007), which provides a foundational framework for understanding how organizations develop and reconfigure information competencies to address rapidly changing environments. Dynamic capabilities represent higher-order capacities that enable firms to sense environmental changes, seize emerging opportunities, and reconfigure resources to maintain competitive fitness (Helfat & Peteraf, 2003). This theoretical perspective is particularly relevant for examining information-intensive ventures where the ability to transform AI-enabled information flows into adaptive capacity becomes crucial for survival and growth (Browder et al., 2024; Cavusgil & Deligonul, 2024). Within this framework, AI innovation capability functions as a dynamic information processing competence that enables ventures to sense market intelligence through advanced analytics, seize knowledge-based opportunities through rapid experimentation, and reconfigure information resources through algorithmic adaptation (Gama & Magistretti, 2025). Strategic resilience, in turn, represents the organizational capacity to anticipate, adapt to, and recover from disruptions while maintaining core functions and capitalizing on emerging opportunities (Browder et al., 2024; Lengnick-Hall et al., 2011). Unlike reactive crisis management, strategic resilience encompasses proactive sensing of environmental threats, agile reconfiguration of resources during disruptions, and transformative learning that positions organizations to emerge stronger from adversity (Engelen et al., 2024; Sahasranamam & Soundararajan, 2022). For information-intensive ventures, resilience manifests through the ability to maintain knowledge continuity, rapidly reconfigure information flows, and leverage disruptions as catalysts for information innovation (Simsek et al., 2024). This conceptualization positions resilience not as a static buffer but as a dynamic, information-driven process through which ventures transform adversity into competitive renewal. Dynamic capability theory thus illuminates how early-stage ventures transform technological information processing potential into organizational knowledge that facilitates continuous adaptation and sustainable performance, providing the theoretical foundation for our hypothesized mediation model.

Existing research on AI-driven information management increasingly highlights the pivotal role of information processing capabilities in detecting environmental changes and reconfiguring operational pathways (Lengnick-Hall et al., 2011; Teece, 2007). While information management literature has established how advanced technologies enhance knowledge creation, streamline information flows, and support market intelligence generation, few studies examine how these information-based innovations complement the adaptive mechanisms underlying organizational resilience in resource-constrained entrepreneurial contexts (Pedota, 2023; Richard et al., 2009). Although dynamic capability theory suggests that information-driven reconfiguration and accelerated knowledge creation form essential processes for sustained entrepreneurial development (Barreto, 2010; Teece et al., 1997), the specific contribution of AI in enhancing these information-intensive capabilities remains underexplored in early-stage ventures with limited information infrastructure (Cillo & Rubera, 2024; Gama & Magistretti, 2025). Significant gaps persist in understanding how AI-enabled information systems simultaneously support exploratory knowledge activities while mitigating informational uncertainties in volatile markets, a challenge that has stimulated renewed interest in information-driven resilience strategies (Browder et al., 2024; Simsek et al., 2024). Current information management literature has not adequately established whether early-stage firms can translate advanced AI-enabled information processing capabilities into sustainable performance outcomes under persistent market uncertainties (Gama & Magistretti, 2025; Nambisan et al., 2017). While numerous studies acknowledge the value of adopting transformative information technologies, significant knowledge gaps remain regarding the mechanisms that connect AI-driven information management to long-term entrepreneurial outcomes (Ameen et al., 2024; Salvato et al., 2020). Consequently, integrating AI innovation capability and strategic resilience within an information management framework represents a particularly timely research direction.

To address these gaps, this study pursues four interrelated research objectives. First, we examine whether strategic resilience fully mediates the relationship between AI innovation capability and entrepreneurial success, testing this mediation model using partial least squares structural equation modeling (PLS-SEM) with two-wave longitudinal data from 357 ventures (RO1). Second, we identify performance gaps through importance-performance map analysis (IPMA) to prioritize managerial interventions for capability development (RO2). Third, recognizing that ventures may achieve success through multiple strategic pathways, we employ fuzzy-set qualitative comparative analysis (fsQCA) to uncover equifinal configurations of AI capabilities and resilience elements that consistently produce high performance (RO3). Fourth, we triangulate these quantitative findings through semi-structured interviews with venture leaders, providing contextualized insights into how these relationships manifest in organizational practice (RO4). This multi-method empirical approach—combining PLS-SEM, IPMA, fsQCA, and qualitative interviews—reflects the theoretical complexity of examining dynamic capabilities in entrepreneurial contexts. The multi-method design is not merely additive; each analytical approach addresses a distinct limitation of the others. PLS-SEM tests linear directional relationships but cannot capture configurational complexity or provide processual insights; fsQCA addresses equifinality and complexity but cannot establish the statistical strength of relational effects; IPMA translates effect sizes into managerial priorities; and qualitative interviews provide the contextual depth to explain why and how the quantitative patterns manifest in organizational practice. Linear modeling techniques illuminate directional relationships central to our mediation hypothesis, while configurational analysis reveals equifinal pathways consistent with dynamic capability theory's emphasis on multiple routes to competitive advantage (Teece, 2007; Ma et al., 2024). Qualitative triangulation provides processual insights into how capabilities evolve and interact in practice, addressing recent calls for richer temporal and processual accounts of capability development (Davidsson et al., 2023; Williams et al., 2021). This methodological pluralism strengthens both internal validity through convergent evidence and external validity through contextual understanding (Jaskiewicz et al., 2015).

To facilitate navigation, Table 1 provides an explicit mapping of each research objective to its corresponding methodological study, analytical tool, and results section.

We use the term information-driven resilience to capture the process through which ventures leverage AI innovation capability—their proficiency in integrating, managing, and applying AI technologies—to build strategic resilience, understood as an adaptive capacity to anticipate, absorb, and transform disruptions into competitive renewal. Our primary contribution lies in developing an information management model that clarifies the fully mediating role of strategic resilience, articulates how AI-driven information initiatives indirectly foster success by first enhancing resilience practices, and provides insights for both theoretical advancement and practical application across diverse information environments. Through a multi-method empirical investigation, this framework demonstrates that technological information processing capabilities alone cannot guarantee success unless organizations simultaneously develop information-based structures and behaviors that enable rapid knowledge creation and application in response to changing conditions. This study addresses these limitations by demonstrating how strategic resilience serves as the essential mediating pathway connecting information-intensive AI capabilities and entrepreneurial success, highlighting that combining technological information processing proficiency with adaptive knowledge management practices can unlock sustainable competitive advantage (Chattopadhyay et al., 2024; Lengnick-Hall et al., 2011).

The remainder of this paper is structured as follows. 2 presents the theoretical background and hypotheses development, drawing upon dynamic capability theory and extant research on AI innovation capability, strategic resilience, and entrepreneurial success. 3 details the research methodology, including the questionnaire design, data collection process, sample characteristics, and analytical techniques employed. Sections 4, 5, 6, and 7 present the results of the PLS-SEM analysis, IPMA, fsQCA, and semi-structured interviews, respectively. 8 discusses the key findings, theoretical implications, and practical insights derived from the study. Finally, 9 outlines the study's theoretical, practical, and policy implications, and 10 concludes by summarizing the main contributions, acknowledging limitations, and outlining avenues for future research.

2. Theoretical background and hypotheses

2.1. *Artificial intelligence innovation capability as information management competence*

Dynamic capability theory provides a foundational framework for understanding how organizations develop and reconfigure internal information competencies to address rapidly changing environments, making it particularly relevant for examining AI's role in early-stage ventures (Teece et al., 1997; Teece, 2007). This theoretical lens emphasizes that sustainable competitive advantage stems not from static resource endowments but from a firm's higher-order ability to integrate, build, and reconfigure information resources in response to emerging opportunities and threats (Barreto, 2010; Helfat & Peteraf, 2003). For information-intensive startups operating in volatile ecosystems, the capacity to transform AI-enabled information flows into adaptive actions is paramount for survival and growth (Cavusgil & Deligonul, 2024; Pedota, 2023). Within this framework, we conceptualize artificial intelligence innovation capability as a critical information-based dynamic capability that enables ventures to sense market intelligence through advanced analytics, seize knowledge-based opportunities through rapid experimentation, and reconfigure information resources through algorithmic adaptation (Carter & Liu, 2025; Gama & Magistretti, 2025).

Artificial intelligence innovation capability represents an organization's proficiency in integrating, managing, and applying artificial intelligence technologies to enhance information processes, knowledge-based decision-making, and information-intensive product/service offerings (Ameen et al., 2024; Gama & Magistretti, 2025). This multidimensional information management construct encompasses three first-order reflective dimensions: artificial intelligence integration innovation, artificial intelligence management innovation, and artificial intelligence application innovation, which collectively form a second-order formative variable (Cillo & Rubera, 2024; Mikalef & Gupta, 2021). Artificial intelligence integration innovation refers to a firm's ability to effectively incorporate artificial intelligence solutions into existing information systems and knowledge workflows, while artificial intelligence management innovation captures the extent to which firms leverage artificial intelligence tools to optimize information resource allocation and enhance knowledge processing efficiency (Ameen et al., 2024). Artificial intelligence application innovation, in turn, reflects a firm's capacity to identify novel information opportunities and develop innovative knowledge-based products or services through the deployment of artificial intelligence technologies (Cillo & Rubera, 2024).

The conceptualization of AI innovation capability as a second-order formative construct is theoretically grounded in the multidimensional nature of organizational capabilities and empirically justified by the distinct yet complementary contributions of its constituent dimensions (Hair et al., 2024; Mikalef & Gupta, 2021). Unlike reflective constructs where indicators are interchangeable manifestations of an underlying latent variable, formative constructs represent combinations of causally independent dimensions that collectively define the higher-order construct (Diamantopoulos & Winklhofer, 2001). In the context of AI innovation capability, the three first-order dimensions—integration, management, and application innovation—represent conceptually distinct facets of how organizations leverage AI technologies. AI integration innovation captures the technical capacity to incorporate AI into existing information infrastructures, AI management innovation reflects the organizational ability to optimize information resource allocation

Table 1
Research Objectives, Methodological Studies, and Result Sections.

Objective	Research Question	Methodological Study	Analytical Tool	Results Section
RO1	Does strategic resilience fully mediate the relationship between AI innovation capability and entrepreneurial success?	Study 1: Capability–Resilience–Performance Pathway	PLS-SEM (two-wave, n = 357)	4
RO2	Which capabilities exhibit performance gaps requiring priority managerial investment?	Study 2: Importance–Performance Mapping	IPMA	5
RO3	What equifinal configurations of AI capabilities and resilience elements produce high entrepreneurial success?	Study 3: Configurational Analysis	fsQCA	6
RO4	How do venture leaders experience and enact information-driven resilience in practice?	Study 4: Processual Qualitative Insights	Semi-structured interviews (n = 10)	7

through AI tools, and AI application innovation embodies the entrepreneurial capacity to identify and exploit novel information opportunities through AI deployment. These dimensions are not merely different measures of the same underlying construct but rather represent unique, non-interchangeable components whose combination determines the overall AI innovation capability of a venture (Ameen et al., 2024; Gama & Magistretti, 2025). A venture might excel in one dimension while developing competencies in others, and each dimension makes a distinct contribution to the firm's overall AI capability. The formative specification acknowledges that these dimensions can vary independently and that their weighted combination—rather than their shared variance—constitutes the higher-order capability construct.

Dynamic capability theory suggests that firms possessing strong artificial intelligence innovation capabilities are better positioned to sense and seize emerging information opportunities, reconfigure knowledge resources, and adapt to changing information environments (Barreto, 2010; Teece et al., 1997). In the context of early-stage ventures operating under high information uncertainty, artificial intelligence innovation capability can serve as a critical enabler of strategic resilience by facilitating rapid knowledge-based decision-making, enhancing information agility, and fostering continuous organizational learning (Ameen et al., 2024; Lengnick-Hall et al., 2011). Artificial intelligence-driven insights can help startups anticipate information shifts, identify latent customer knowledge needs, and proactively adjust their strategic information direction, thereby strengthening their capacity to withstand and recover from information disruptions (Browder et al., 2024; Simsek et al., 2024). Moreover, the integration of artificial intelligence technologies into core information processes can streamline knowledge flows, reduce information response times, and improve knowledge resource utilization, further bolstering a venture's resilience in the face of information adversity (Gama & Magistretti, 2025; Pedota, 2023). Consequently, we propose the following hypothesis:

H1. Artificial intelligence innovation capability positively influences strategic resilience in entrepreneurial ventures.

2.2. *Strategic resilience as knowledge adaptation capacity*

Strategic resilience represents an organization's capacity to anticipate, adapt to, and recover from information disruptions while maintaining core knowledge functions and capitalizing on emerging information opportunities (Browder et al., 2024; Lengnick-Hall et al., 2011). Distinct from passive robustness or reactive crisis management, strategic resilience is a proactive and dynamic capability that involves cognitive foresight, behavioral agility, and contextual embeddedness to navigate and even thrive amidst adversity (Williams et al., 2021). It enables ventures not only to "bounce back" to their original state but also to "bounce forward" by learning from disruptions and reconfiguring resources to emerge stronger and more competitive (Ren et al., 2024; Simsek et al., 2024).

Strategic resilience is conceptualized as a second-order formative construct comprising cognitive, behavioral, and contextual elements that represent distinct yet synergistic mechanisms through which organizations achieve adaptive capacity (Browder et al., 2024; Lengnick-Hall et al., 2011). This formative specification is theoretically appropriate because resilience emerges from the combined presence of these heterogeneous capabilities rather than representing their common underlying factor (Hair et al., 2024). Cognitive elements capture mental models and sense-making processes that enable anticipation and opportunity identification during disruptions. Behavioral elements reflect action-oriented capabilities involving strategic adjustment, operational continuity, and initiative implementation. Contextual elements encompass relational and resource-based capacities including stakeholder engagement and environmental adaptation. These three dimensions operate through different organizational mechanisms—cognitive processing, behavioral execution, and contextual navigation—and a firm's overall resilience is determined by how these distinct capabilities combine rather than by a common resilience factor that manifests through all three (Engelen et al., 2024). A venture could possess strong cognitive resilience (clear vision, opportunity identification) while still developing behavioral resilience (rapid strategic adjustment), demonstrating that these dimensions are not interchangeable reflections of a single latent construct but rather independent contributors whose integration constitutes strategic resilience. The formative approach appropriately models this relationship wherein the dimensions form resilience rather than being caused by it.

Cognitive elements capture a firm's ability to maintain clear information vision and quickly identify potential knowledge opportunities during information crises (Seyfi et al., 2025). Behavioral elements reflect a firm's capacity to rapidly adjust information strategies, maintain effective knowledge operations, and successfully implement new information initiatives in the face of disruptions (Peretz-Andersson et al., 2024; Ren et al., 2024). Contextual elements encompass a firm's proficiency in maintaining strong information stakeholder relationships, effectively utilizing available knowledge resources, and successfully adapting to changing information environments (Breidbach et al., 2024).

Dynamic capability theory suggests that firms possessing strong strategic resilience are better positioned to withstand information adversity, recover from knowledge setbacks, and achieve superior information-driven performance outcomes in uncertain environments (Barreto, 2010; Teece et al., 1997). In the context of entrepreneurial ventures operating under high information uncertainty, strategic resilience can serve as a critical enabler of entrepreneurial success by facilitating rapid knowledge adaptation, fostering information innovation, and enhancing competitive information positioning (Sahasranamam & Soundararajan, 2022; Scholz et al., 2020). Resilient ventures demonstrate the ability to maintain clear strategic information direction, quickly mobilize knowledge resources, and effectively engage with information stakeholders, thereby increasing their chances of survival and growth in the face of information adversity (Li et al., 2023; Xie & Wu, 2022). Moreover, the capacity to proactively identify and seize emerging information opportunities amid disruptions can lead to the development of new knowledge products, information services, or knowledge-based business models, further bolstering a venture's competitive advantage and entrepreneurial success (Hora & Dutta, 2013; Toorajipour et al., 2024; Venkatesh et al., 2017). Consequently, we propose the following hypothesis:

H2. Strategic resilience positively influences entrepreneurial success in entrepreneurial ventures.

2.3. *Information-driven entrepreneurial success*

Entrepreneurial success represents the extent to which new ventures achieve their desired information-driven performance outcomes, including knowledge-enabled revenue growth, information-based market share expansion, and customer information base development (Richard et al., 2009; Staniewski et al., 2025). This multidimensional information management construct captures the ability of early-stage firms to translate their innovative information capabilities into tangible knowledge advantages and superior information-driven market positioning (Davidsson et al., 2023; Venkatesh et al., 2017). Dynamic capability theory suggests that entrepreneurial success emerges from the effective orchestration of information resources and knowledge competencies to address rapidly evolving information environments (Barreto, 2010; Teece et al., 1997). In the context of information-intensive startups, artificial intelligence innovation capability represents a critical information resource that can enhance entrepreneurial success by enabling firms to develop novel information products, optimize knowledge processes, and identify untapped information market opportunities (Ameen et al., 2024; Gama & Magistretti, 2025). However, the mere possession of artificial intelligence capabilities may not guarantee success unless firms simultaneously cultivate strategic resilience—the capacity to anticipate, adapt to, and recover from information disruptions while maintaining core knowledge functions (Browder et al., 2024; Lengnick-Hall et al., 2011). Strategic resilience enables ventures to effectively leverage their artificial intelligence capabilities by facilitating rapid knowledge-based decision-making, fostering information agility, and promoting continuous learning in the face of uncertainty (Engelen et al., 2024; Simsek et al., 2024). Therefore, we hypothesize:

H3. Strategic resilience mediates the relationship between artificial intelligence innovation capability and entrepreneurial success in entrepreneurial ventures.

Company age and size reflect information resource availability and knowledge experience that may affect innovation adoption patterns (Browder et al., 2024; Lengnick-Hall et al., 2011; Roppelt et al., 2025), while industry type and region capture institutional information variations that could shape technological implementation and resilience development (Ameen et al., 2024; Simsek et al., 2024). Fig. 1 presents the research model.

3. Methods

3.1. Methodology flowchart

The research flowchart illustrates a systematic seven-stage process for examining the relationships between artificial intelligence innovation capability, strategic resilience, and entrepreneurial success. The process begins with questionnaire development through expert panel validation and pre-testing, followed by data collection from Chinese and European startups, bias and robustness analyses, PLS-SEM modeling, importance-performance mapping, fuzzy-set qualitative comparative analysis, and semi-structured interviews to triangulate the quantitative findings. Fig. 2 presents the detailed methodological framework that guides this research.

3.2. Research context

Artificial intelligence innovation capability represents a critical strategic resource for entrepreneurial ventures in today's digital economy. The research context focuses on early-stage startups (≤ 5 years old) in China and Europe that have actively implemented artificial intelligence technologies in their operations and products/services (Ameen et al., 2024; Gama & Magistretti, 2025). These startups operate in an environment characterized by rapid technological change, high uncertainty, and intense competition, where strategic resilience becomes essential for survival and success (Browder et al., 2024; Engelen et al., 2024). The selection of China and Europe as research settings provides complementary institutional contexts. While China represents an emerging economy with strong government support for artificial intelligence development and entrepreneurship, Europe offers a mature market environment with established regulatory frameworks for technology adoption (Zahra et al., 2024). This dual-context approach enables examination of how artificial intelligence innovation capability and strategic resilience manifest across different institutional environments (Cavusgil & Deligonul, 2024). The focus on early-stage startups is particularly relevant as these ventures face heightened vulnerability during their initial years while simultaneously possessing the flexibility to integrate novel technologies and adapt their business models (Chattopadhyay et al., 2024). The sample spans multiple industries including information technology, healthcare, manufacturing, and financial services, allowing for broader generalizability while controlling for industry-specific effects (Simsek et al., 2024).

To provide a richer contextual foundation for interpreting our findings, we present here a systematic overview of the AI technologies deployed and the organizational functions covered within our sample of 357 early-stage ventures.

AI Technology Profile: Among the 357 ventures in our sample, the most widely adopted AI technologies included machine learning platforms (deployed by 68.3% of ventures), natural language processing tools (54.9%), computer vision applications (38.7%), and robotic process automation (31.4%). A smaller but growing segment (19.1%) had implemented large language models (LLMs) or generative AI tools for product development or customer interaction.

Organizational Functions Covered by AI: AI implementation spanned multiple functional domains. The most commonly reported AI application areas were: product/service development and innovation (74.2% of ventures), customer service and engagement (62.3%), marketing analytics and demand forecasting (58.5%), operations and supply chain optimization (44.8%), financial management and risk assessment (39.2%), and human resource management (22.7%). This functional diversity reflects the broad scope of AI integration innovation captured in our three-dimensional construct.

Industry-Specific AI Adoption Patterns: Significant variation in AI function adoption existed across industries. Information technology ventures (35.2% of sample) predominantly applied AI to product development and R&D acceleration. Healthcare ventures (16%) focused primarily on diagnostic systems, patient data analytics, and workflow automation. Financial services firms (14.6%) concentrated AI deployment on risk assessment, fraud detection, and algorithmic investment tools. Manufacturing ventures (12.6%) emphasized supply chain optimization and quality control. E-commerce ventures (8.1%) primarily utilized AI for marketing analytics, recommendation systems, and personalized customer experiences. Education and training ventures (10.4%) focused on content generation, adaptive learning systems, and student engagement analytics.

AI Maturity and Resource Investment: Ventures reported having dedicated between 8% and 24% of their annual operating budgets to AI development and integration, with a mean allocation of 14.7%. Approximately 43.1% of ventures had established a dedicated AI team or AI function within their organizational structure, while the remaining 56.9% embedded AI capabilities within existing functional teams. This variation in AI governance structure corresponds meaningfully to our AI management innovation dimension and has implications for the behavioral resilience mechanisms we observe.

This research setting provides an ideal context for examining how artificial intelligence innovation capability influences entrepreneurial success through the mediating role of strategic resilience.

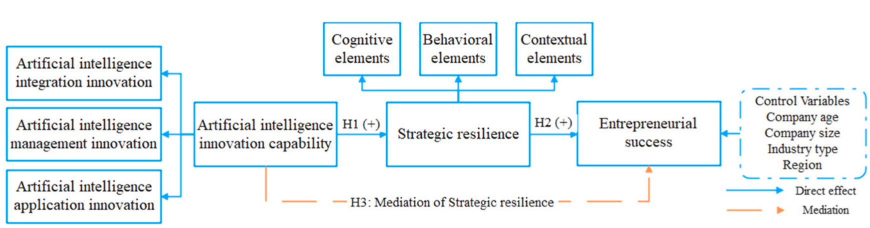

Fig. 1. Research model.

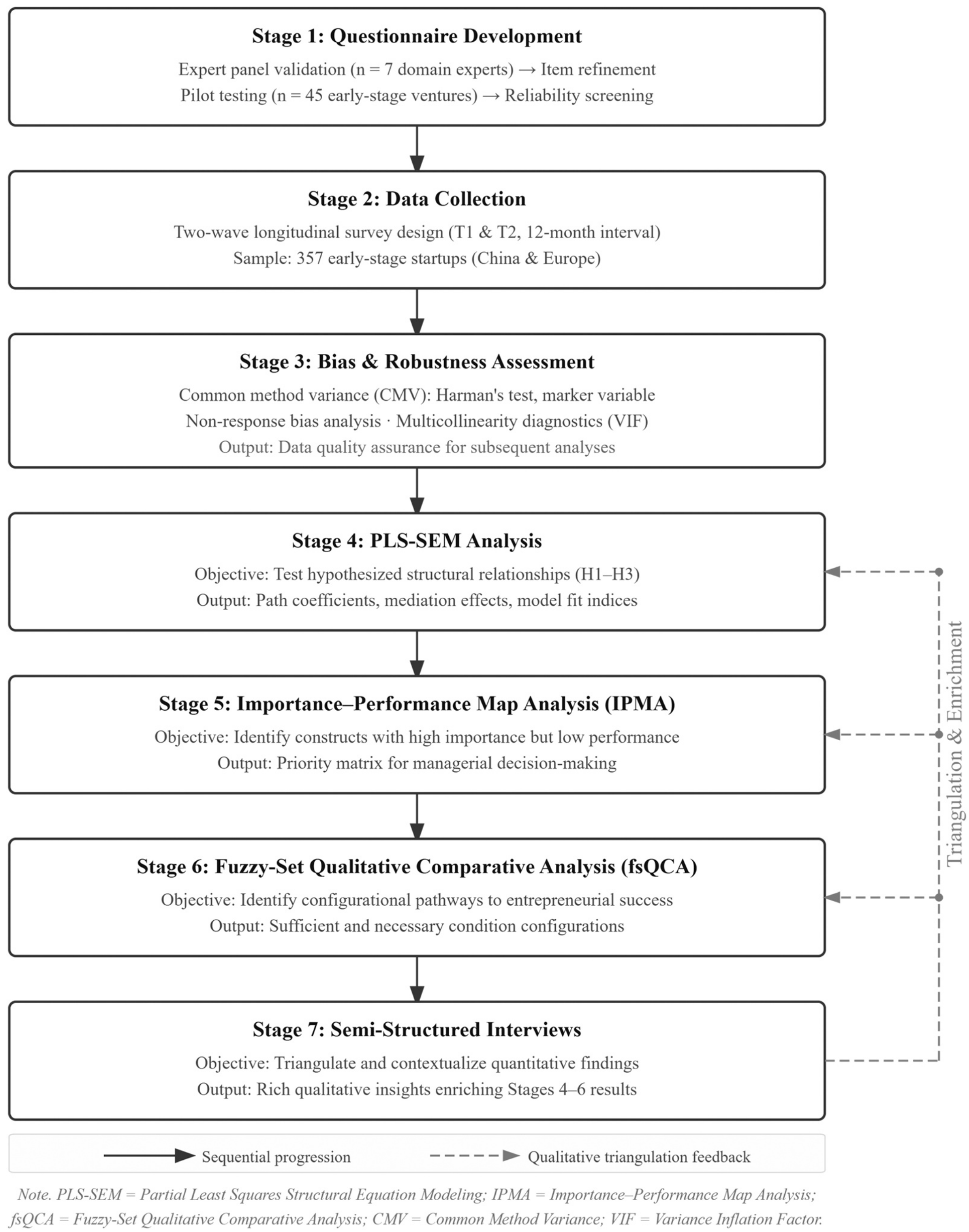

Fig. 2. Methodology flowchart.

3.3. *Questionnaire design*

The measurement instrument underwent a rigorous multi-stage development to ensure reliability, validity, and contextual alignment with early-stage startups operating under intense uncertainty (Gama & Magistretti, 2025; Pinky et al., 2024). The process commenced with a comprehensive review and synthesis of existing scales, including the assessment of managerial and organizational constructs in entrepreneurial contexts (Richard et al., 2009; Staniewski et al., 2025), culminating in an initial item pool tailored to the present research aims. Subsequently, this pool was refined through multiple expert panel consultations, bringing together scholars and industrial practitioners with specialized backgrounds in strategic management, digital innovation, and startup incubation (Browder et al., 2024; Lengnick-Hall et al., 2011). These experts examined the clarity, coverage, and theoretical alignment of individual measurement items, which led to a second iteration of the questionnaire that was more specific to artificial intelligence innovation, strategic resilience, and entrepreneurial success dimensions (Ameen et al., 2024). The next phase comprised a pilot test with experienced respondents drawn from various industries, including information technology, healthcare, and financial services, thereby gathering critical feedback on item comprehensibility and response accuracy (Xie & Wu, 2022). This phase verified the internal consistency of the instrument (Cronbach's alpha > 0.70), and revealed minor wording ambiguities that were rectified in the subsequent version. Afterward, a back-translation procedure was employed to confirm cultural equivalence between English and Chinese/European language versions, minimizing interpretational biases across diverse environmental settings (Simsek et al., 2024). Entrepreneurial success (Table 2) is measured using objective data from the past year. The selection of one-year growth metrics is theoretically and empirically justified for our research context. For early-stage ventures ($\leq$ 5 years old) operating in

Table 2
Questionnaire.

Questions		References
Artificial intelligence integration innovation	Our company effectively incorporates artificial intelligence solutions into existing business processes. Our company successfully adapts artificial intelligence technologies to meet specific business needs. Our company efficiently combines artificial intelligence technologies with current information technology systems. Our company regularly updates our artificial intelligence applications based on business requirements.	(Ameen et al., 2024; Gama & Magistretti, 2025; Nambisan et al., 2017)
Artificial intelligence management innovation	Our company actively utilizes artificial intelligence tools in managerial decision-making. Our company optimizes resource management processes through artificial intelligence. Our company leverages artificial intelligence technologies to enhance management efficiency.	
Artificial intelligence application innovation	Our company identifies new business opportunities through artificial intelligence applications. Our company creates innovative solutions using existing artificial intelligence technologies. Our company continuously improves existing products or services using artificial intelligence. Our company introduces novel artificial intelligence-enabled services/products to meet customer needs.	
Cognitive elements	Our company maintains a clear vision during challenging times. Our management team quickly identifies potential opportunities in crises. Our company demonstrates strong problem-solving capabilities when facing difficulties.	(Browder et al., 2024; Lengnick-Hall et al., 2011; Seyfi et al., 2025)
Behavioral elements	Our company rapidly adjusts strategies in response to environmental changes. Our company maintains effective operations during disruptions. Our company successfully implements new initiatives during uncertain times.	
Contextual elements	Our company maintains strong relationships with key stakeholders during difficulties. Our company effectively utilizes available resources in challenging situations. Our company successfully adapts to changing business environments.	
Entrepreneurial success	Annual revenue growth rate in the past year (%) Market share growth in the past year (%) Customer base growth in the past year (%)	(Richard et al., 2009; Staniewski et al., 2025)

high-velocity, technology-intensive environments, one-year performance indicators offer several advantages over multi-year retrospective measures (Davidsson et al., 2023; Richard et al., 2009). First, these metrics align with the temporal dynamics of strategic resilience—an organizational capacity that enables rapid sensing and response to environmental disruptions within quarters to one-year cycles rather than extended multi-year periods (Lengnick-Hall et al., 2011). Second, for resource-constrained startups, traditional profitability metrics can be misleading due to deliberate strategies prioritizing rapid scaling and market penetration over near-term profit generation (Staniewski et al., 2025). Growth-oriented indicators—revenue expansion, market share gains, and customer base development—serve as more accurate proxies for venture viability and competitive positioning in entrepreneurial contexts. Third, these three metrics collectively provide a holistic, multidimensional assessment of a venture's ability to create value and secure competitive advantage, capturing economic performance (revenue growth), relative market positioning (market share growth), and future potential (customer base growth). Fourth, our two-wave data collection design with one-year temporal separation between AI capability/resilience measurement (Wave 1) and success measurement (Wave 2) addresses recency bias concerns by establishing temporal precedence and ensuring that performance outcomes reflect post-capability development results rather than pre-existing trajectories. Finally, the objective nature of these metrics—based on actual financial and operational records rather than perceptual assessments—provides additional protection against retrospective bias. This measurement approach aligns with best practices in entrepreneurial performance research (Richard et al., 2009) while appropriately capturing the dynamic adaptation processes central to our theoretical framework. Other items used a seven-point Likert-type scale (1 = strongly disagree; 7 = strongly agree) to capture nuanced perceptions of artificial intelligence-oriented activities and resilience behaviors (Richard et al., 2009).

3.4. Data collection and sample

Purposive sampling technique was employed to identify and recruit early-stage startups actively implementing artificial intelligence technologies in China and Europe. This sampling approach enables targeted selection of information-rich cases while ensuring theoretical relevance and data quality (Browder et al., 2024; Sahasranamam & Soundararajan, 2022). The data collection process spanned 14 months, utilizing both online questionnaires and follow-up interviews to gather comprehensive insights into artificial intelligence innovation capabilities and strategic resilience.

The sampling frame comprised startups identified through entrepreneurship associations, technology incubators, and artificial intelligence industry consortia in both regions. Two screening criteria were applied: (1) company age less than 5 years, and (2) active implementation of artificial intelligence in operations or products/services. This dual-filter approach helped ensure sample homogeneity while capturing variations in artificial intelligence adoption patterns (Chattopadhyay et al., 2024; Gama & Magistretti, 2025). Data collection proceeded in two waves to minimize common method bias and establish temporal precedence between predictors and outcomes. The first wave gathered information on artificial intelligence innovation capability and strategic resilience as our focal independent and mediating variables, respectively. The second wave, conducted one year later, collected data on entrepreneurial success as our dependent variable, alongside firm characteristics as control variables. This temporal separation helps reduce potential spurious correlations and addresses concerns about reverse causality (Salvato et al., 2020; Williams et al., 2021).

Our two-wave design with single-point predictor measurement aligns with dynamic capability theory, which posits that organizational capabilities developed at baseline enable future adaptation and performance (Barreto, 2010; Teece et al., 1997). Measuring AI capability and strategic resilience at Time 1 and entrepreneurial success at Time 2 directly captures this temporal dynamic—how baseline capabilities influence subsequent performance outcomes. Re-measuring predictors at Wave 2 would fundamentally alter our research question and potentially confound capability evolution with outcome achievement, particularly problematic in our context of rapidly developing early-stage ventures (Davidsson et al., 2023; Williams et al., 2021). This design follows established precedents in entrepreneurship research examining capability-performance relationships (Hora & Dutta, 2013; Salvato et al., 2020; Venkatesh et al., 2017).

The final sample consisted of 357 valid responses (191 Chinese startups and 166 European startups), representing a 51.0% initial response rate and 89.0% valid response rate. Sample size adequacy was confirmed through both G*Power analysis (minimum required n = 98) and the ten-times rule (minimum required n = 230) (Hu et al., 2025). Table 3 presents the sample characteristics, revealing a balanced distribution across company age, size, industry type, and geographical region. This diverse yet focused sample enables robust examination of the hypothesized relationships while controlling for potential confounding effects (Simsek et al., 2024; Zahra et al., 2024). The one-year measurement window for entrepreneurial success is particularly appropriate for our sample characteristics. Early-stage ventures in our study operate under compressed decision cycles and face existential threats that require near-term performance signals for adaptive learning and resource allocation (Salvato et al., 2020; Williams et al., 2021). Multi-year retrospective measures would introduce severe survivorship bias, as many early-stage ventures do not survive beyond their initial years due to market volatility, resource constraints, or strategic pivots, making multi-year longitudinal data unavailable or unreliable for a substantial portion of the target population. Furthermore, extended measurement periods would introduce confounding effects from multiple intervening environmental changes, making it difficult to attribute long-term outcomes to AI capabilities and resilience measured at baseline. Our methodological choice represents an optimal balance between capturing meaningful performance outcomes and maintaining sample representativeness and analytical rigor in the entrepreneurial context (Browder et al., 2024; Chattopadhyay et al., 2024).

Table 3
Sample characteristics.

Characteristics		%
Company Age	1 year	5.9
	2 years	24.6
	3 years	38.4
	4 years	19.3
	5 years	11.8
Company Size	1-10 employees	21.6
	11-50 employees	52.7
	51-200 employees	19.3
	201 + employees	6.4
Industry Type	Information Technology	35.2
	Healthcare	16
	Manufacturing	12.6
	Financial Services	14.6
	Education & Training	10.4
	E-commerce	8.1
	Other	3.1
Region	China	53.5
	Europe	46.5

To address potential concerns about retrospective bias, our design incorporates multiple protective features. First, Wave 1 measurements of AI capability and strategic resilience employed forward-looking, present-tense items assessing current organizational capabilities rather than requiring retrospective recall. Second, Wave 2 measurement of entrepreneurial success relies on objective performance metrics (revenue growth rate, market share growth, customer base growth) based on actual financial and operational records rather than perceptual assessments, substantially reducing retrospective distortion risks (Richard et al., 2009). Third, the one-year temporal separation provides sufficient time for capabilities to manifest in observable outcomes while maintaining theoretical relevance in high-velocity entrepreneurial contexts (Staniewski et al., 2025; Xie & Wu, 2022). This temporal window aligns with best practices in entrepreneurship research examining capability-performance relationships where baseline capabilities predict subsequent performance without requiring repeated predictor measurement.

To complement and triangulate our quantitative findings, we conducted semi-structured interviews with a purposively selected subset of our survey respondents. Following the completion of the two-wave survey, we invited founders, CEOs, and CTOs from ventures that represented diverse configurations of AI capabilities and resilience profiles to participate. This purposive sampling strategy was designed to gather rich, context-specific insights into the mechanisms underlying our quantitative results (Jaskiewicz et al., 2015). A total of 10 interviews were conducted between November 2024 and January 2025, with each interview lasting approximately 45–60 min. The interviews were conducted via secure video conferencing platforms to accommodate the geographical distribution of participants across China and Europe. We developed a semi-structured interview protocol based on our theoretical framework (see Appendix A), with open-ended questions designed to explore how leaders perceive the role of AI in building strategic resilience, how they navigate environmental disruptions, and the specific practices that connect AI initiatives to performance outcomes. This format provided a consistent thematic guide while allowing the flexibility to probe emergent insights and unique experiences (Williams et al., 2021). All interviews were recorded with participant consent, transcribed verbatim, and analyzed using thematic analysis. This involved a systematic process of open coding to identify initial concepts, followed by axial coding to group these concepts into broader themes, and finally, selective coding to integrate these themes into a coherent narrative that explains the processual dynamics of information-driven resilience.

Obtaining two-wave longitudinal survey data from early-stage ventures represents a particularly demanding methodological undertaking. Early-stage ventures are inherently unstable—many do not survive the 12-month window between Wave 1 and Wave 2 data collection—making longitudinal sampling from this population substantially more challenging than cross-sectional approaches. Our 89.0% valid response rate at Wave 2 and our 14-month data collection period, spanning both Chinese and European entrepreneurial ecosystems through entrepreneurship associations, technology incubators, and AI industry consortia, required intensive follow-up protocols and relationship-based access that is difficult to replicate. This investment in data quality produces a unique dataset that simultaneously addresses temporal precedence concerns, captures cross-institutional variation, and maintains the sample homogeneity required for meaningful comparison—a combination rarely achieved in AI and entrepreneurship research.

3.5. Analytical techniques

3.5.1. Overview of multi-method research strategy

Our study employs a multi-method research design, integrating four distinct analytical techniques to provide a comprehensive, robust, and nuanced examination of the relationships between artificial intelligence innovation capability, strategic resilience, and entrepreneurial success. Each stage of our analysis serves a specific objective, and their combination allows us to triangulate findings and generate deeper insights than a single method could afford (Jaskiewicz et al., 2015). Our analytical strategy follows a sequential, complementary design where each study builds upon the insights of the previous one.

Our research design requires four analytical methods because our theoretical questions operate at four distinct levels of inference that are irreducible to each other. First, we employ PLS-SEM (Study 1) because our core theoretical contribution is a mediation hypothesis specifying that strategic resilience intervenes in the AI capability–entrepreneurial success relationship. Testing this directional, probabilistic relationship requires a method capable of simultaneously estimating structural paths, calculating indirect effects, and assessing predictive relevance across a sample of sufficient statistical power—capabilities that PLS-SEM provides for models with complex formative second-order constructs (Hair et al., 2022; Hair et al., 2024). Second, we employ IPMA (Study 2) because PLS-SEM path coefficients provide information about the statistical strength of relationships but do not directly answer the managerial question of where performance investments will generate the greatest returns. IPMA uniquely maps the importance of each construct (its total effect on the outcome) against its current performance level (its mean score), revealing actionable priority gaps that are invisible in standard SEM output. Third, we employ fsQCA (Study 3) because PLS-SEM and IPMA both operate under symmetric, net-effect assumptions that preclude examination of how specific combinations of conditions jointly produce outcomes. In real entrepreneurial settings, capabilities substitute for each other, interact in non-additive ways, and multiple distinct pathways can lead to the same outcome (equifinality)—phenomena that only set-theoretic, configurational methods can capture (Fiss, 2011; Teece, 2007). Fourth, we employ semi-structured interviews (Study 4) because quantitative methods, however powerful, cannot reveal the processual dynamics and organizational sense-making through which capabilities are enacted in real organizational contexts. The interviews provide essential interpretive depth that explains why the quantitative patterns arise and how they manifest in practice, addressing recent calls for richer processual accounts of capability development (Davidsson et al., 2023; Williams et al., 2021). Together, these four methods form a sequential, complementary design where each study addresses the limitations of the previous one, building toward a comprehensive and convergently validated understanding of the AI capability–resilience–success nexus.

All PLS-SEM analyses were conducted using SmartPLS 4.1 (Hair et al., 2024), employing 5000 bootstrap samples for significance testing and confidence interval construction. fsQCA was conducted using fsQCA 4.1 software (Ragin & Davey, 2023). Thematic analysis of interview data was supported by NVivo 20, following the systematic open-axial-selective coding protocol outlined in 3.5.4.

3.5.2. Study 1: PLS-SEM rationale and design

Study 1: Partial Least Squares Structural Equation Modeling (PLS-SEM) serves as the first and most critical stage of our analysis. Its primary objective is to test our core theoretical hypotheses (H1-H3), specifically examining the linear, directional, and mediating relationships proposed in our research model. We selected PLS-SEM, a variance-based approach, for several compelling reasons that align with our research goals and data structure. First, our model includes second-order formative constructs (AI innovation capability and strategic resilience), for which PLS-SEM is widely recognized as the superior analytical choice compared to covariance-based SEM (CB-SEM) (Hair et al., 2022; Jarvis et al., 2003). Second, the primary goal of our initial analysis is prediction—understanding how AI innovation capability and strategic resilience predict entrepreneurial success. PLS-SEM is optimized for maximizing the explained variance of endogenous constructs and is thus ideally suited for predictive research models (Hair et al., 2024). This variance-based approach is therefore ideal for validating the proposed mediation pathway and assessing the predictive power of our model before proceeding to more complex, exploratory analyses.

3.5.3. Study 2: IPMA rationale and design

Study 2: Importance-Performance Map Analysis (IPMA) logically follows the PLS-SEM. It extends the structural model results by graphically mapping the relative importance and performance of the constructs, allowing us to identify which specific capabilities have the greatest relative impact on driving entrepreneurial success. This translates our statistical findings into actionable managerial priorities.

3.5.4. Studies 3 and 4: fsQCA and interview rationale

Study 3: fuzzy-set Qualitative Comparative Analysis (fsQCA) moves beyond the net-effect, linear assumptions of SEM. Having established the foundational model in Study 1, we use fsQCA to uncover complex, configurational pathways to success. This method allows us to explore equifinality—the principle that different combinations of AI capabilities and resilience elements can lead to the same successful outcome—which is a central tenet of dynamic capability theory that cannot be captured by SEM alone (Fiss, 2011; Teece, 2007).

Study 4: Semi-Structured Interviews provide the final layer of analysis. We use qualitative data to triangulate and provide rich contextual depth to our quantitative findings, offering processual insights into how venture leaders perceive and experience the development of information-driven resilience in practice.

This methodological pluralism, with PLS-SEM as the foundational stage, ensures both the statistical validity of our core theoretical model and a deep, contextually-grounded understanding of the phenomenon under investigation. Multiple analytical techniques are employed to ensure robust examination of the hypothesized relationships between artificial intelligence innovation capability, strategic resilience, and entrepreneurial success. PLS-SEM serves as the primary analytical tool due to its capability to handle complex path models and formative constructs in emerging research contexts (Galindo-Martin et al., 2025). The selection of PLS-SEM is particularly appropriate given the formative nature of artificial intelligence innovation capability as a second-order construct and the exploratory nature of examining strategic resilience in digital transformation contexts (Al Halbusi et al., 2025). To complement the variance-based analysis, IPMA is utilized to identify key dimensions that significantly impact entrepreneurial success while being currently underdeveloped (Hair et al., 2024). Additionally, fsQCA is employed to uncover complex configurational patterns that may lead to high entrepreneurial success (Li et al., 2025), acknowledging the potential equifinality in achieving desired outcomes (Fiss, 2011).

The decision to specify AI innovation capability and strategic resilience as second-order formative rather than reflective constructs warrants explicit justification, particularly in the context of early-stage ventures with developing organizational capabilities (Hair et al., 2024; Jarvis et al., 2003). This decision was guided by four theoretical and empirical considerations. First, the direction of causality between the construct and its indicators: in formative models, indicators cause the construct, whereas in reflective models, the construct causes the indicators. For both AI innovation capability and strategic resilience, theory suggests that organizational possession of the constituent dimensions (integration, management, application for AI capability; cognitive, behavioral, contextual for resilience) produces the higher-order capability rather than the higher-order capability manifesting through these dimensions. A venture develops AI innovation capability by building competencies in integration, management, and application—these capabilities are the building blocks that create overall AI capability, not symptoms of an underlying AI capability factor. Second, interchangeability of indicators: reflective indicators should be highly correlated and interchangeable because they measure the same construct, whereas formative indicators should be relatively independent because they represent distinct facets. Our theoretical framework and empirical evidence (low VIF values, modest correlations between first-order dimensions) demonstrate that the dimensions are distinct rather than interchangeable, supporting formative specification. Third, construct meaning and validity: in reflective models, dropping an indicator does not fundamentally change the construct's meaning if other indicators remain; in formative models, eliminating an indicator removes part of the construct's conceptual domain. For our constructs, omitting any dimension (e.g., measuring AI capability without application innovation, or resilience without cognitive elements) would fundamentally alter and diminish what the construct represents, supporting the formative approach. Fourth, contextual appropriateness for early-stage ventures: while early-stage ventures may have limited data maturity in some respects, the formative approach is particularly well-suited to this context because it acknowledges heterogeneity in capability development trajectories. Early-stage ventures often develop capabilities unevenly—some may excel in AI integration while still building application capabilities—and the formative model appropriately captures this developmental reality by allowing dimensions to vary independently rather than assuming they must move in lockstep as a reflective model would require (Ameen et al., 2024; Mikalef & Gupta, 2021). This developmental heterogeneity strengthens rather than undermines the rationale for formative specification in our research context.

3.6. Common method bias and robustness

Common method bias poses potential threats to the validity of research findings in behavioral studies. Multiple ex-ante and ex-post procedures were implemented to address this concern. The ex-ante procedures included temporal separation of predictor and criterion variables through a two-wave data collection process spanning one year, with artificial intelligence innovation capability and strategic resilience measured in the first wave, and entrepreneurial success measured in the second wave. The two-wave design establishes temporal precedence between AI capability/resilience measurement and entrepreneurial success outcomes, providing evidence consistent with a causal interpretation while acknowledging that observational designs cannot eliminate all potential confounders (Podsakoff et al., 2003). Our design captures the theoretically appropriate temporal relationship where baseline capabilities (Time 1) influence subsequent performance outcomes (Time 2), consistent with dynamic capability theory's emphasis on how organizational capabilities developed at one point enable future adaptation and performance (Lengnick-Hall et al., 2011; Teece et al., 1997). This single-point predictor measurement approach avoids the conceptual ambiguity that would arise from re-measuring evolving capabilities at multiple time points in our rapidly developing early-stage venture sample, where such re-measurement would risk confounding capability evolution with outcome achievement (Davidsson et al., 2023). The ex-post assessment began with Harman's single-factor test, which revealed that the first factor accounted for 38.739% of total variance, well below the 50% threshold, indicating no substantial common method bias. To provide more rigorous evidence, a marker variable technique was employed using "perceived ease of use" as the marker variable. The marker variable demonstrated satisfactory psychometric properties (Cronbach's $\alpha = 0.816$, $CR = 0.869$, $AVE = 0.728$) and showed no significant correlations with the study variables ($p > 0.05$), further confirming that common method bias was not a serious concern (Li et al., 2025).

Non-response bias was assessed through early-late response comparison, following established procedures in entrepreneurship research (Armstrong & Overton, 1977). The sample was split into early and late respondents based on response timing. Independent *t*-tests revealed no significant differences between early and late respondents across key organizational characteristics, including company size ($p = 0.149$). These results indicate no systematic differences between the two groups, suggesting that non-response bias was not a problematic issue in this study (Al Halbusi et al., 2025).

Multicollinearity diagnostics were performed using variance inflation factors (VIF). The analysis revealed indicator VIF values ranging from 1.680 to 2.515, substantially below the conservative threshold of 5.0 suggested in the literature (Hair et al., 2022). The mean VIF of 1.975 further indicates that multicollinearity does not threaten the stability of parameter estimates. Additionally, the Gaussian copula approach implemented through SmartPLS 4.1 with 5000 bootstrap samples showed no significant endogeneity concerns ($p > 0.05$), supporting the robustness of the structural model estimates (Hair et al., 2024).

4. Study 1: validating the capability-resilience-performance pathway

4.1. Measurement model

The measurement model (Table 4 and Table 5) demonstrates strong psychometric properties across all constructs (Hair et al., 2022). Factor loadings for individual items range from 0.830 to 0.900, exceeding the threshold of 0.7, indicating robust item reliability. Internal consistency reliability is established through Cronbach's alpha values (0.805-0.886) and composite reliability scores (0.807-0.886), both surpassing the 0.7 criterion. Average variance extracted (AVE) values (0.720-0.799) satisfy the 0.5 threshold, confirming convergent validity. Discriminant validity is assessed through three complementary approaches. The Fornell-Larcker criterion shows that the square root of each construct's AVE (diagonal values: 0.848-0.894) exceeds its correlations with other constructs. Heterotrait-monotrait (HTMT) ratios fall below 0.85, ranging from 0.329 to 0.776. Cross-loadings reveal that each item loads highest on its intended construct. These results collectively establish discriminant validity among constructs. For the formative second-order constructs, rigorous assessment procedures were employed to evaluate indicator selection, weighting, and specification appropriateness (Diamantopoulos & Siguaw, 2006; Hair et al., 2024). First, theoretical justification for each indicator's inclusion was established through our conceptual framework (Sections 2.2 and 2.3), ensuring that each first-order dimension captures a distinct facet of the higher-order construct that cannot be omitted without fundamentally altering the construct's meaning. Second, multicollinearity among formative indicators was assessed through variance inflation factors (VIF), which range from 1.203 to 2.204, well below the conservative threshold of 5.0, indicating that the dimensions are sufficiently distinct to contribute unique information to the higher-order construct (Hair et al., 2022). This low multicollinearity provides evidence consistent with the claim that the dimensions are not redundant manifestations of the same underlying factor—a key requirement for formative specification. Third, all formative indicators demonstrate statistically significant and substantial weights, with significance assessed using bias-corrected accelerated (BCa) bootstrapping with 5000 bootstrap samples, following recommendations by Hair et al. (2024) for formative construct validation. All weights achieved statistical significance at $p < 0.001$, with 95% bootstrap confidence intervals excluding zero for all six first-order indicators (AIAI: weight = 0.385, 95% BCI [0.375, 0.398]; AIII: weight = 0.374, 95% BCI [0.365, 0.385]; AIMI: weight = 0.382, 95% BCI [0.370, 0.395]; BE: weight = 0.445, 95% BCI [0.422, 0.472]; GE: weight = 0.353, 95% BCI [0.324, 0.376]; NE: weight = 0.449, 95% BCI [0.426, 0.476]). This bootstrapping procedure provides evidence consistent with the claim that each dimension makes a statistically meaningful and non-trivial contribution to its respective higher-order construct. The significant weights validate our theoretical assertion that all three dimensions are essential components of the overall capability, as eliminating any dimension would fundamentally diminish the construct's conceptual breadth and predictive validity. Fourth, the magnitude of the weights reveals relatively balanced contributions across dimensions (weights differ by less than 0.1), suggesting that no single dimension dominates the higher-order construct—a pattern consistent with our theoretical conceptualization of these as complementary rather than

Table 4
Reliability and convergent validity.

Construct	Items	Loadings > 0.7	VIF	Cronbach's Alpha > 0.7	Composite reliability > 0.7	AVE > 0.5	Mean	SD
Artificial intelligence application innovation (AIAI)	AIAI1	0.865	2.372	0.884	0.884	0.743	4.614	1.678
	AIAI2	0.858	2.235					
	AIAI3	0.877	2.515					
	AIAI4	0.847	2.091					
Artificial intelligence integration innovation (AIII)	AIII1	0.870	2.402	0.886	0.886	0.745	4.823	1.542
	AIII2	0.870	2.380					
	AIII3	0.864	2.328					
	AIII4	0.849	2.129					
Artificial intelligence management innovation (AIMI)	AIMI1	0.888	2.261	0.868	0.868	0.792	4.485	1.652
	AIMI2	0.886	2.210					
	AIMI3	0.895	2.361					
Behavioral elements (BE)	BE1	0.860	1.926	0.841	0.842	0.758	4.592	1.675
	BE2	0.866	1.948					
	BE3	0.886	2.119					
Entrepreneurial success (ES)	ES1	0.894	2.318	0.874	0.874	0.799	4.387	1.728
	ES2	0.900	2.469					
	ES3	0.887	2.277					
Cognitive elements (GE)	GE1	0.866	1.881	0.805	0.807	0.720	4.768	1.621
	GE2	0.848	1.707					
	GE3	0.830	1.680					
Contextual elements (NE)	NE1	0.879	2.073	0.836	0.837	0.753	4.421	1.697
	NE2	0.880	2.083					
	NE3	0.845	1.780					

Construct	Items	VIF	Weight	95% BCI	P value
Artificial intelligence innovation capability	AIAI	2.081	0.385	[0.375, 0.398]	< 0.001
	AIII	1.941	0.374	[0.365, 0.385]	< 0.001
	AIMI	2.204	0.382	[0.370, 0.395]	< 0.001
Strategic resilience	BE	1.634	0.445	[0.422, 0.472]	< 0.001
	GE	1.203	0.353	[0.324, 0.376]	< 0.001
	NE	1.678	0.449	[0.426, 0.476]	< 0.001

Table 5
Fornell-Larcker criterion, heterotrait-monotrait ratio and cross loadings.

Construct	AIAI	AIII	AIMI	BE	ES	GE	NE
AIAI	**0.862**	0.706	0.776	0.582	0.615	0.461	0.606
AIII	0.625	**0.863**	0.744	0.548	0.573	0.399	0.528
AIMI	0.680	0.652	**0.890**	0.556	0.603	0.393	0.485
BE	0.502	0.473	0.475	**0.871**	0.653	0.427	0.729
ES	0.542	0.505	0.526	0.560	**0.894**	0.512	0.671
GE	0.389	0.337	0.329	0.352	0.429	**0.848**	0.466
NE	0.521	0.455	0.414	0.612	0.575	0.383	**0.868**

Note: Bolded numerals represent the square roots of the average variance extracted. The values positioned in the lower-left region of the table, beneath the diagonal, serve as the Fornell–Larcker criterion, while the figures positioned in the upper-right region above the diagonal represent the HTMT ratios.

Construct	AIAI	AIII	AIMI	BE	ES	GE	NE
AIAI1	**0.865**	0.544	0.564	0.440	0.472	0.298	0.447
AIAI2	**0.858**	0.516	0.584	0.417	0.453	0.305	0.466
AIAI3	**0.877**	0.517	0.604	0.418	0.463	0.310	0.421
AIAI4	**0.847**	0.577	0.591	0.454	0.479	0.427	0.464
AIII1	0.543	**0.870**	0.537	0.404	0.417	0.290	0.370
AIII2	0.537	**0.870**	0.563	0.386	0.431	0.296	0.372
AIII3	0.532	**0.864**	0.576	0.448	0.441	0.246	0.447
AIII4	0.545	**0.849**	0.576	0.396	0.453	0.331	0.381
AIMI1	0.605	0.567	**0.888**	0.411	0.455	0.251	0.330
AIMI2	0.613	0.586	**0.886**	0.449	0.494	0.334	0.417
AIMI3	0.597	0.588	**0.895**	0.406	0.454	0.293	0.358
BE1	0.445	0.389	0.447	**0.860**	0.477	0.302	0.495
BE2	0.440	0.415	0.398	**0.866**	0.496	0.303	0.529
BE3	0.427	0.432	0.398	**0.886**	0.489	0.314	0.573
ES1	0.520	0.463	0.494	0.495	**0.894**	0.412	0.530
ES2	0.490	0.449	0.468	0.486	**0.900**	0.379	0.526
ES3	0.440	0.440	0.447	0.520	**0.887**	0.359	0.483
GE1	0.321	0.277	0.272	0.284	0.338	**0.866**	0.334
GE2	0.347	0.308	0.289	0.330	0.358	**0.848**	0.338
GE3	0.322	0.272	0.277	0.280	0.399	**0.830**	0.301
NE1	0.406	0.403	0.349	0.538	0.539	0.334	**0.879**
NE2	0.522	0.426	0.389	0.550	0.521	0.342	**0.880**
NE3	0.428	0.354	0.340	0.505	0.434	0.321	**0.845**

Note: Boldface values indicate factor loadings for the construct indicators.

redundant facets. This empirical pattern supports the formative specification by demonstrating that the construct emerges from the combination of distinct components rather than representing a common underlying factor. Finally, nomological validity was assessed by examining whether the formatively specified constructs demonstrate theoretically expected relationships with other constructs in the model, which is confirmed by the strong and significant structural paths ($\beta = 0.624$ for H1, $\beta = 0.656$ for H2) and high explanatory power ($R^2 = 39.0\%$ and 43.8%).

4.2. *Structural model and hypothesis testing*

The structural model (Fig. 3 and Table 6) demonstrates strong support for all hypothesized relationships (Hair et al., 2022). Artificial intelligence innovation capability exhibits a significant positive effect on strategic resilience ($\beta = 0.624$, $p < 0.001$, $f^2 = 0.637$), providing robust support for H1. The substantial effect size ($f^2 > 0.35$) indicates that artificial intelligence innovation capability plays a crucial role in developing strategic resilience among entrepreneurial ventures. Strategic resilience demonstrates a strong positive influence on entrepreneurial success ($\beta = 0.656$, $p < 0.001$, $f^2 = 0.760$), supporting H2. The large effect size suggests that strategic resilience serves as a vital capability for achieving entrepreneurial success in technology-driven startups. This finding is consistent with dynamic capability theory, highlighting how firms leverage adaptive capacities to achieve superior performance outcomes. In line with the theory's emphasis on capability building over time, the results are consistent with the hypothesized temporal ordering whereby AI innovation capability precedes and facilitates the development of strategic resilience (Barreto, 2010; Teece et al., 1997). However, as our evidence comes from an observational two-wave design, alternative explanations cannot be fully ruled out; we therefore use cautious language when discussing causality. The mediation analysis reveals that strategic resilience significantly mediates the relationship between artificial intelligence innovation capability and

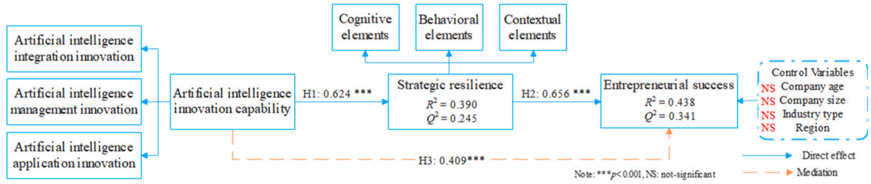

Fig. 3. PLS-SEM model.

Table 6
Hypothesis test.

Hypotheses	β	SD	95% BCI [LL, UL]	p	f^2	Support?
H1: Artificial intelligence innovation capability ➔ Strategic resilience (+)	0.624	0.034	[0.555, 0.687]	< 0.001	0.637	√
H2: Strategic resilience ➔ Entrepreneurial success (+)	0.656	0.029	[0.599, 0.711]	< 0.001	0.760	√
H3: Strategic resilience mediates the relationship between artificial intelligence innovation capability and entrepreneurial success.	0.409	0.031	[0.345, 0.473]	< 0.001	-	√

Note: BCa = Bias-Corrected and Accelerated; LL = Lower Limit; UL = Upper Limit; based on 5000 bootstrap samples.

entrepreneurial success (β = 0.409, $p < 0.001$), supporting H3. This mediation effect underscores how artificial intelligence innovation capability enhances entrepreneurial success through the development of strategic resilience, rather than through direct effects alone. The path coefficients and significance levels remain stable after controlling for company age, size, and industry type. Notably, the control for region (China vs. Europe) was not significant, indicating that the core model relationships are robust across these distinct institutional environments.

4.3. Model Fit, R^2, Q^2, f^2

The structural model demonstrates robust predictive power and explanatory capability across multiple assessment criteria (Hair et al., 2022; Hair et al., 2024). The standardized root mean square residual (SRMR) value of 0.069 falls well below the conservative threshold of 0.08, indicating strong model fit. The coefficient of determination (R^2) reveals that artificial intelligence innovation capability explains 39.0% of the variance in strategic resilience, while the model accounts for 43.8% of the variance in entrepreneurial success, suggesting substantial explanatory power in the context of early-stage technology ventures. Consistent with PLS-SEM's prediction-oriented logic, our model evaluation prioritizes both explanatory power (R^2) and predictive relevance (Q^2) rather than solely relying on model fit indices appropriate to covariance-based approaches (Hair et al., 2022; Hair et al., 2024). The R^2 values of 0.390 for strategic resilience and 0.438 for entrepreneurial success demonstrate substantial explanatory power for early-stage venture contexts, where unexplained variance from unobserved industry-specific factors is expected. The Stone-Geisser Q^2 values of 0.245 and 0.341, respectively, both substantially exceed the zero threshold, confirming that the model possesses meaningful out-of-sample predictive relevance—the primary criterion in variance-based structural modeling. This dual focus on R^2 and Q^2 ensures that our model is not merely fitting the observed data but generating insights with predictive utility beyond the estimation sample.

The Stone-Geisser Q^2 values derived through blindfolding procedures with an omission distance of 7 exceed zero for both strategic resilience (0.245) and entrepreneurial success (0.341), demonstrating the model's predictive relevance. The effect sizes (f^2) for the relationships between artificial intelligence innovation capability and strategic resilience (0.637), and between strategic resilience and entrepreneurial success (0.760), both surpass the threshold for large effects (0.35), indicating substantial practical significance of these relationships in the entrepreneurial context. These results collectively validate the model's theoretical foundation while establishing its practical utility for understanding how artificial intelligence capabilities contribute to entrepreneurial outcomes through strategic resilience mechanisms.

5. Study 2: IPMA results

Importance-performance map analysis reveals distinct patterns in the relative influence of artificial intelligence innovation capability and strategic resilience on entrepreneurial success (Hair et al., 2022; Hair et al., 2024). Strategic resilience demonstrates higher importance (0.836) compared to artificial intelligence innovation capability (0.561), while both constructs show moderate performance levels (58.934 and 60.987 respectively). This finding suggests (Fig. 4) that while startups have developed reasonable levels of both capabilities, strategic resilience serves as a more critical driver of entrepreneurial success in uncertain environments. The higher performance score for artificial intelligence innovation capability indicates that startups have made progress in technological implementation, yet the lower importance coefficient suggests these technological capabilities alone may not be sufficient for success. The analysis points to strategic resilience as a key area for managerial attention, given its high importance but relatively lower performance score. These results align with dynamic capability theory by highlighting how the transformation of technological resources into adaptive capabilities ultimately drives entrepreneurial outcomes.

6. Study 3: fsQCA results

This study expands on the preceding PLS-SEM results by incorporating fuzzy-set qualitative comparative analysis (fsQCA) to capture asymmetric, equifinal, multifaceted, and conjunctive causal relationships in predicting entrepreneurial success (Li et al., 2025). This complementary approach offers deeper insights into how artificial intelligence innovation capability intersects with resilience factors to generate favorable outcomes.

The first step calibrates each predictor and outcome into fuzzy-set membership scores using the direct method (Ragin, 2008). We set the qualitative anchors for full membership, crossover, and full non-membership at the 90th percentile, the scale midpoint, and the 10th percentile of each variable’s distribution, respectively, consistent with fsQCA practice for Likert-type organizational data (Fiss, 2011). These anchors correspond to membership scores of 0.95, 0.50, and 0.05. Raw scores were then transformed into fuzzy-set memberships bounded

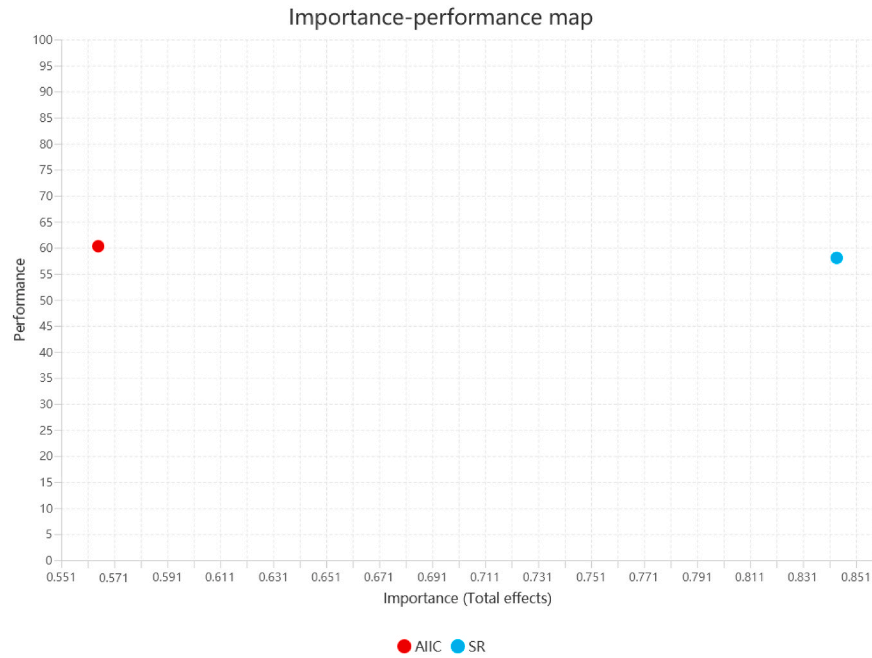

Fig. 4. IPMA of entrepreneurial success.

between 0 and 1 based on these anchors. Necessity analysis indicated that no single condition reaches a consistency above 0.90, suggesting that no single condition is necessary for entrepreneurial success (Fiss, 2011). We then constructed a truth table for entrepreneurial success, setting the minimum frequency threshold to two cases, the raw consistency threshold to 0.80, and the proportional reduction in inconsistency (PRI) threshold to 0.70. Configurations meeting both frequency and consistency criteria were retained for subsequent solution derivation.

The fsQCA dataset yields complex, parsimonious, and intermediate solutions, from which the intermediate solution is selected for interpretability while retaining completeness (Fiss, 2011). This solution reveals six configurations (see Table 7) that consistently lead to high entrepreneurial success, as indicated by consistency values exceeding 0.90. The overall solution coverage is 0.633, and the solution consistency is 0.861, suggesting these solution paths demonstrate strong explanatory power and empirical relevance. To enhance the practical utility and theoretical interpretability of these findings, we provide detailed interpretations of each configuration below, explicating their distinct theoretical meanings and actionable implications (Bawack et al., 2025).

Table 7
Configurations for high entrepreneurial success (fsQCA intermediate solution).

Construct	Configurations					
	1	2	3	4	5	6
AIII		◉	◉	◉	◉	◉
AIMI	◉			◉	●	◉
AIAI			◉		◉	●
GE	◉	◉	◉	◉		◉
BE	◉	◉			●	◉
NE	◉	◉	◉	◉	◉	
Raw coverage	0.429	0.434	0.440	0.434	0.453	0.431
Unique coverage	0.032	0.014	0.008	0.008	0.073	0.050
Consistency	0.905	0.907	0.910	0.921	0.925	0.924
Solution coverage	0.633					
Solution consistency	0.861					

Note:◉ = core condition present; ●= peripheral condition present; blank cells = 'don't care' conditions (neither presence nor absence is required for the outcome). Raw coverage indicates the proportion of cases with high entrepreneurial success covered by each configuration. Unique coverage indicates the proportion exclusively explained by each configuration. Consistency indicates the degree to which the configuration is a subset of high entrepreneurial success. All configurations are derived from the intermediate solution of the fsQCA analysis. These configurations describe sufficient conditions for high entrepreneurial success observed among high-performing ventures in our sample and should not be interpreted as universal prescriptions.

Configuration 1 (Management-Centric Resilience Pathway): This configuration demonstrates that ventures can achieve high entrepreneurial success by emphasizing AI management innovation (AIMI) as their primary AI capability, combined with comprehensive strategic resilience across all three dimensions—cognitive elements (GE), behavioral elements (BE), and contextual elements (NE). This pathway suggests that ventures focusing on optimizing resource management processes through AI tools—such as intelligent resource allocation, automated decision support systems, and AI-enhanced operational efficiency—can succeed even when their AI integration or application innovations are not fully developed. The presence of all three resilience dimensions indicates that these ventures leverage AI-enabled management efficiency to build holistic adaptive capacity. Practically, this configuration is particularly relevant for resource-constrained startups that may lack the technical infrastructure for deep AI integration but can deploy AI management tools to enhance organizational agility and decision-making speed during disruptions.

Configuration 2 (Integration-Centric Resilience Pathway): This pathway reveals that AI integration innovation (AIII)—the ability to effectively incorporate AI solutions into existing business processes and information systems—serves as a sufficient condition for success when combined with full strategic resilience capabilities (GE, BE, NE). Ventures following this pathway excel at embedding AI technologies within their operational workflows, ensuring seamless technology adoption across organizational functions. The core presence of cognitive, behavioral, and contextual resilience elements suggests these ventures translate their strong AI integration foundation into comprehensive adaptive capabilities. This configuration is especially applicable to ventures operating in established industries where successful AI adoption requires careful integration with legacy systems and organizational routines, rather than radical application innovations. Managers in such contexts should prioritize building organizational capabilities for smooth technology assimilation and change management while simultaneously developing resilience mechanisms to navigate integration challenges.

Configuration 3 (Integration-Application Synergy Pathway): This configuration demonstrates that the combination of AI integration innovation (AIII) and AI application innovation (AIAI)—without requiring strong AI management innovation—can drive entrepreneurial success when cognitive elements (GE) and contextual elements (NE) are present. This pathway suggests that ventures capable of both incorporating AI into existing systems (integration) and identifying novel business opportunities through AI deployment (application) can succeed even with less developed AI management practices, provided they maintain clear strategic vision (cognitive elements) and strong stakeholder relationships (contextual elements). Notably, behavioral resilience is not a core requirement in this configuration, suggesting that strategic foresight and external relationship management capabilities compensate for potential deficiencies in rapid strategic adjustment. This configuration is particularly relevant for technology-intensive startups in emerging AI application domains, where the ability to quickly deploy AI for new use cases while maintaining integration coherence drives competitive advantage. Practitioners should focus on building dual capabilities in AI technology adoption and opportunity identification while ensuring leadership maintains strategic clarity and stakeholder engagement.

Configuration 4 (Balanced AI Capability Pathway): This pathway shows that ventures combining AI integration innovation (AIII) and AI management innovation (AIMI)—creating a balanced AI capability foundation—can achieve success even without strong application innovation, provided cognitive elements (GE) and contextual elements (NE) are present. This configuration suggests that operational excellence in AI adoption (integration) and efficiency optimization (management) creates sufficient value when the organization maintains strategic clarity and environmental adaptation capabilities. The absence of application innovation as a core condition indicates that incremental improvements to existing processes through AI, rather than radical new applications, can drive success in certain contexts. This finding challenges the common assumption that AI-driven innovation must always involve novel product or service offerings. Practically, this configuration applies to ventures in mature industries where competitive advantage stems from operational efficiency and cost management rather than product differentiation. Managers should focus on perfecting AI integration across workflows and leveraging AI for management optimization while building cognitive capabilities to anticipate market changes and contextual capabilities to maintain resource access during disruptions.

Configuration 5 (Application-Driven Contextual Adaptation Pathway): This configuration reveals a distinctive pathway where AI application innovation (AIAI)—the capacity to identify new business opportunities and develop innovative products/services through AI—becomes the central driver of success when combined with strong contextual resilience (NE), with AI management innovation (AIMI) and behavioral elements (BE) playing supportive peripheral roles, while AI integration innovation (AIII) serves as an additional core enabling condition. Notably, cognitive resilience is absent as a core condition, suggesting that for application-focused ventures, the ability to quickly deploy AI-enabled innovations and maintain strong external resource networks can compensate for potential gaps in strategic foresight or internal problem-solving capabilities. This configuration is particularly relevant for entrepreneurial ventures pursuing disruptive innovation strategies in rapidly evolving markets, where speed of innovation deployment and stakeholder support matter more than comprehensive strategic planning. Practitioners following this pathway should prioritize building capabilities for rapid AI-enabled product development, maintaining strong relationships with customers, investors, and partners, while ensuring sufficient (though not necessarily exceptional) integration and management capabilities to operationalize innovations.

Configuration 6 (Cognitive-Behavioral Excellence Pathway): This pathway demonstrates that ventures emphasizing cognitive elements (GE) and behavioral elements (BE)—maintaining clear strategic vision, rapid opportunity identification, quick strategic adjustment, and effective implementation—can achieve success through balanced AI capabilities (core AIII and AIMI, peripheral AIAI) even when contextual resilience is not strongly developed. This configuration suggests that internal organizational capabilities for sensing, seizing, and reconfiguring resources can substitute for external relationship-based resilience mechanisms. The pattern indicates that ventures with strong leadership vision and agile execution capabilities can leverage their AI integration and management foundations to navigate disruptions through proactive internal adaptation rather than relying primarily on stakeholder support. This configuration is especially applicable to ventures led by experienced entrepreneurial teams with strong domain expertise and decision-making capabilities, operating in contexts where rapid internal pivoting provides competitive advantage. Managers should focus on building robust strategic sensing mechanisms, fostering organizational agility, and developing comprehensive AI integration and management capabilities while recognizing that stakeholder relationship management, though important, need not be the primary resilience mechanism.

Equifinality and Strategic Implications: The emergence of six distinct configurations demonstrates the principle of equifinality—multiple pathways can lead to the same outcome of high entrepreneurial success, and no single configuration is universally necessary or sufficient (Fiss, 2011). This finding has profound theoretical and practical implications. Theoretically, it challenges linear, deterministic models suggesting that specific AI capabilities or resilience elements are universally critical for entrepreneurial success. Instead, our results reveal that ventures can strategically combine different AI innovation capabilities (integration, management, application) with different resilience dimensions (cognitive, behavioral, contextual) to achieve equifinal outcomes, reflecting the complex, context-dependent nature of capability orchestration in resource-constrained entrepreneurial settings (Browder et al., 2024; Williams et al., 2021).

Practically, the equifinality of these pathways offers entrepreneurs and managers strategic flexibility in capability development. Ventures need not pursue excellence across all AI capability dimensions simultaneously, nor must they develop all resilience elements uniformly. Rather, they can identify which configuration aligns best with their founding team strengths, industry characteristics, resource constraints, and competitive positioning. For instance, ventures with strong technical founders but limited industry connections might pursue Configuration 3 (integration-application synergy) or Configuration 6 (cognitive-behavioral excellence), emphasizing internal capabilities over external relationship management. Conversely, ventures led by experienced industry entrepreneurs with established networks might follow Configuration 5 (application-driven contextual adaptation), leveraging stakeholder relationships to compensate for less developed internal strategic foresight. This strategic selectivity is particularly valuable for resource-constrained startups that cannot simultaneously invest in all capability dimensions and must make deliberate choices about where to concentrate limited resources.

Furthermore, the configurational patterns reveal important substitution and complementarity effects among conditions. Configurations 1, 2, and 4 all emphasize comprehensive resilience (presence of cognitive, behavioral, and contextual elements) combined with specific AI capabilities, suggesting that when ventures build holistic resilience, they can succeed with narrower AI capability profiles. In contrast, Configuration 5 shows that ventures emphasizing a specific AI capability (application innovation) with strong contextual resilience can succeed despite gaps in cognitive resilience, indicating that external resource access can partially substitute for internal strategic foresight. These insights enable managers to understand not only what capabilities to build but also how different capabilities interact to create value, facilitating more nuanced strategic planning. The diversity of pathways also has implications for policy and ecosystem support, suggesting that entrepreneurship support programs should provide differentiated assistance tailored to ventures' chosen capability configurations rather than adopting one-size-fits-all approaches.

To provide a complete set-theoretic analysis, Table 8 presents the configurations for the negated outcome (low entrepreneurial success). This analysis employed the same calibration anchors and truth table specifications as the high-success analysis, with the outcome variable negated. The analysis reveals five configurations for low entrepreneurial success, with a solution coverage of 0.612 and solution consistency of 0.914. Crucially, asymmetric analysis reveals that the configurations producing low success are not mirror images of those producing high success—a finding consistent with set-theoretic logic (Fiss, 2011). Specifically, the low-success configurations are dominated by a generalized capability deficiency, in which resilience elements are largely missing (GE, BE, and NE are absent in most configurations (four out of five)) and AI innovation capability is underdeveloped (AIAI is absent in all configurations, with AIII and AIMI frequently absent). Rather than finding configurations where AI capabilities are strongly present but fail due to a lack of resilience, the asymmetric analysis reveals that venture failure in this context stems from a fundamental organizational poverty—where early-stage startups fail to develop both the technological tools and the adaptive capacities simultaneously. This highlights the asymmetric nature of capability development: while there are multiple equifinal pathways to succeed (as seen in the high-success configurations), the pathway to low success is predominantly driven by a generalized failure to accumulate both information processing (AI) and organizational adaptation (resilience) resources.

7. Study 4: interview results

The interviews confirm that artificial intelligence innovation has become a catalyst for strengthening strategic resilience in early-stage companies facing resource constraints and market volatility (Cillo & Rubera, 2024). Multiple interviewees emphasize that deploying artificial intelligence solutions encourages rapid adjustments in product development, operational workflows, and risk monitoring, thereby aligning with the quantitative findings regarding the positive link between artificial intelligence innovation capability and strategic resilience. These interview insights underscore the importance of organizational adaptability as a crucial channel through which artificial intelligence solutions contribute to entrepreneurial performance (Hora & Dutta, 2013). The outcome of these observations creates a robust basis for interpreting how artificial intelligence-enabled initiatives support growth trajectories in uncertain environments (Chattopadhyay et al., 2024).

The interview accounts highlight that the bridging role of strategic resilience emerges most visibly when leaders proactively leverage artificial intelligence to accelerate organizational learning and cross-functional collaboration. As indicated by Interviewee 1 and Interviewee 7 in Table 9, active experimentation in product development fosters a receptive culture for timely decision-making, which ultimately enhances market responsiveness. Interviewee 2 expresses a similar perspective by noting that process automation not only reduces operational costs but also creates agility to experiment with novel service offerings, thereby strengthening resilience. Such remarks strongly reinforce the quantitative finding that strategic resilience mediates the relationship between artificial intelligence innovation capability and entrepreneurial success. These experiences provide coherent evidence that robust resilience structures can amplify the benefits gained from artificial intelligence-oriented initiatives.

The interviewees further stress that short-term gains from artificial intelligence projects require continuous refinement of strategic approaches in order to maintain momentum in dynamic competitive contexts. Interviewee 4 and Interviewee 10 notably discuss how artificial intelligence applications in manufacturing, such as supply chain optimization and quality control, need ongoing adjustments to sustain positive effects on customer satisfaction and profitability. These data points parallel the structural model results indicating that strategic resilience is crucial for turning initial artificial intelligence-driven achievements into sustained outcomes. This alignment highlights the idea that resilience serves as an essential integrative force that bolsters growth even across diverse industrial environments.

Taken together, the interview evidence yields three foundational processual insights that directly complement and deepen our quantitative findings. First, AI innovation capability and strategic resilience are not developed in isolation but co-evolve through iterative cycles of experimentation and organizational learning, with AI tools serving as accelerants for the sensing and seizing processes that resilience enables. Second, the value of AI emerges specifically through organizational interpretation and translation processes—resilience provides the interpretive infrastructure through which AI-generated patterns become embedded in strategic decisions and operational routines. Third, the sustainability of initial AI-driven performance gains depends critically on leaders' commitment to continuous strategic refinement, underscoring that resilience is not a one-time organizational achievement but an ongoing adaptive practice. These processual insights form the empirical foundation for the integrated discussion in 8, where we synthesize across all four studies to develop our four key theoretical contributions.

Table 8
Configurations for low entrepreneurial success (fsQCA intermediate solution).

Construct	Configurations				
	1	2	3	4	5
AIII	○	◎	○	◎	
AIMI	◎	○	◎		◎
AIAI	◎	○	◎	◎	◎
GE	◎	◎		◎	◎
BE	◎		◎	◎	◎
NE		◎	◎	◎	◎
Raw coverage	0.478	0.494	0.508	0.491	0.488
Unique coverage	0.016	0.032	0.046	0.029	0.026
Consistency	0.947	0.949	0.938	0.951	0.954
Solution coverage	0.612				
Solution consistency	0.914				

Note:◎ = core condition absent; ○ = peripheral condition absent; blank cells = 'don't care' conditions (neither presence nor absence is required for the outcome). Raw coverage indicates the proportion of cases with low entrepreneurial success covered by each configuration. Unique coverage indicates the proportion exclusively explained by each configuration. Consistency indicates the degree to which the configuration is a subset of low entrepreneurial success. All configurations are derived from the intermediate solution of the fsQCA analysis. These configurations describe sufficient conditions for low entrepreneurial success observed among low-performing ventures in our sample and should not be interpreted as universal prescriptions.

8. Discussion

The overarching theoretical contribution of this study is the empirical demonstration that strategic resilience fully mediates the relationship between AI innovation capability and entrepreneurial success—a finding that fundamentally challenges the technology-performance determinism prevalent across information management, digital transformation, and AI innovation literatures (Nambisan et al., 2017; Piccoli et al., 2024). Prior scholarship on AI and competitive advantage has frequently treated technological capability as a proximate driver of performance, implicitly assuming that advanced information processing translates directly into superior outcomes. Our multi-method evidence reveals, with convergent support from PLS-SEM (indirect effect $\beta = 0.409$, $p < 0.001$), IPMA (strategic resilience importance = 0.836 vs. AI capability importance = 0.561), fsQCA (six equifinal resilience-anchored configurations), and qualitative interviews (resilience as active transformation mechanism), that this assumption systematically misidentifies where the real performance leverage resides. The value of AI is not direct; it is unlocked by first building the

Table 9
Interviewee characteristics.

ID	Position	Industry	Company age	Company size	AI implementation area
I1	CEO	Information Technology	3 years	11-50	Product Development
I2	CTO	Healthcare	2 years	51-200	Process Automation
I3	Founder	Financial Services	4 years	11-50	Customer Service
I4	COO	Manufacturing	3 years	51-200	Supply Chain
I5	CEO	E-commerce	2 years	11-50	Marketing Analytics
I6	CTO	Education & Training	3 years	11-50	Content Generation
I7	Founder	Information Technology	4 years	1-10	Product Innovation
I8	CEO	Healthcare	2 years	11-50	Diagnostic Systems
I9	COO	Financial Services	3 years	11-50	Risk Assessment
I10	CTO	Manufacturing	2 years	11-50	Quality Control

organizational capacity for adaptive learning and agile knowledge reconfiguration. Our multi-method investigation, visually represented across Figs. 1–5, demonstrates theoretical coherence in examining how AI innovation capability influences entrepreneurial success through strategic resilience.

Fig. 1 establishes our theoretical model grounded in dynamic capability theory, proposing three core hypotheses regarding the relationships among AI innovation capability, strategic resilience, and entrepreneurial success. Fig. 2 illustrates how our seven-stage methodology systematically addresses these hypotheses through complementary analytical approaches—questionnaire development, data collection, bias assessment, PLS-SEM, IPMA, fsQCA, and interviews—each contributing unique insights to our understanding of the phenomenon. Fig. 3 presents PLS-SEM results confirming the mediation model (indirect effect $\beta = 0.409$, $p < 0.001$), demonstrating that AI innovation capability influences entrepreneurial success exclusively through strategic resilience rather than through direct pathways. Fig. 4 extends these findings through importance-performance map analysis, revealing that while strategic resilience demonstrates higher importance for entrepreneurial success (total effects = 0.836) compared to AI innovation capability (total effects = 0.561), it exhibits lower performance (58.934 vs. 60.987), suggesting a critical performance gap requiring managerial attention. Together, these figures provide converging evidence that the value of AI-enabled information processing capabilities is unlocked by first building organizational adaptive capacity, challenging linear technology-performance assumptions prevalent in information management literature (Browder et al., 2024; Nambisan et al., 2017). Fig. 5 synthesizes these multi-method findings into an integrative conceptual summary, illustrating how the core PLS-SEM mediation model (inner layer), the IPMA importance-performance gap (middle layer), the six equifinal fsQCA configurational pathways (outer layer), and the interview-derived processual dynamics (feedback loop overlay) collectively converge to demonstrate that AI value is unlocked through strategic resilience.

We organize our discussion around four core findings that challenge and extend existing theoretical perspectives on AI innovation, strategic

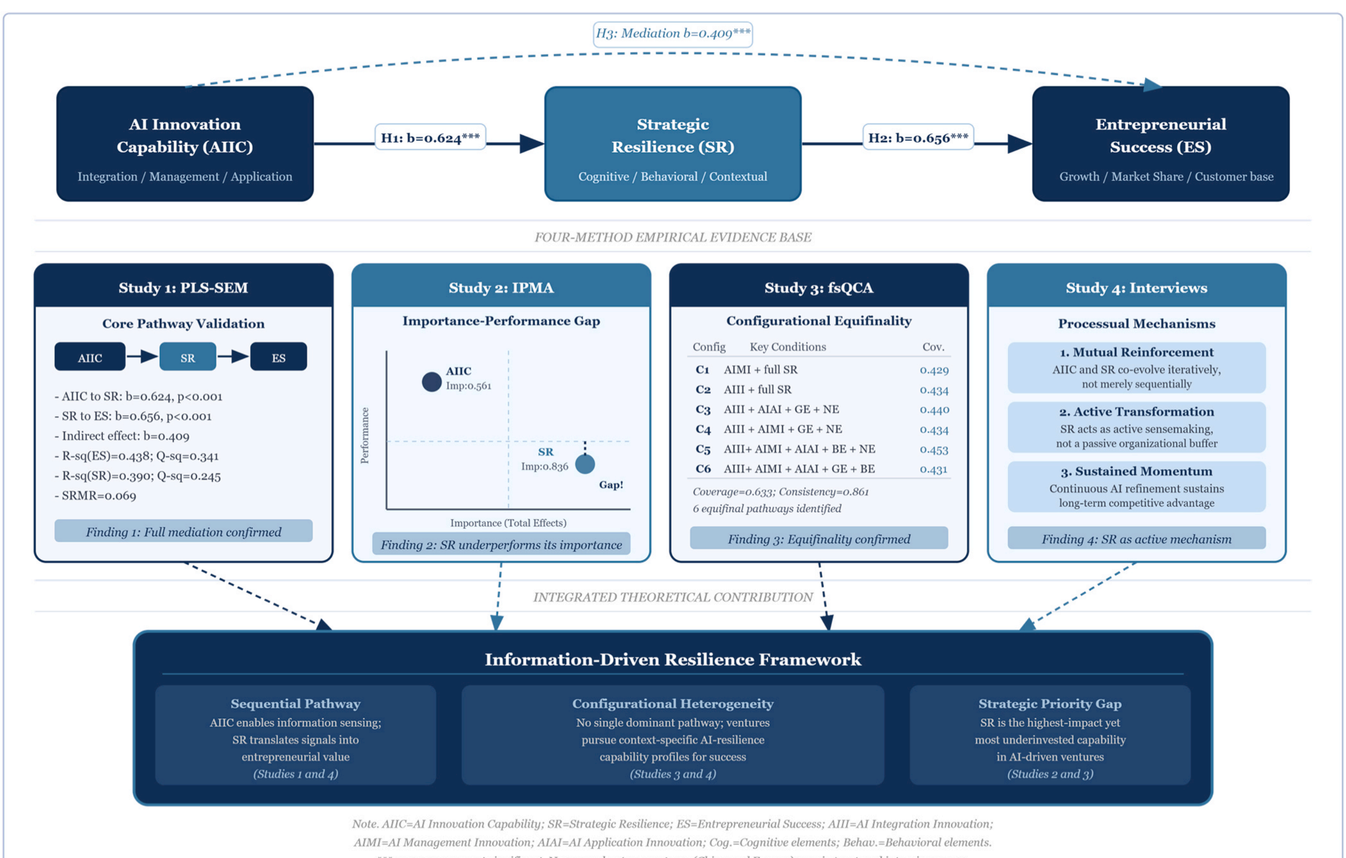

Fig. 5. Multi-method synthesis: Converging evidence on information-driven resilience.

resilience, and entrepreneurial success. Each finding represents a distinct theoretical contribution with implications for how scholars conceptualize technology-enabled adaptation in resource-constrained entrepreneurial contexts.

8.1. Finding 1: mediation—AI value requires resilience intermediation

Our PLS-SEM results reveal that strategic resilience fully mediates the relationship between AI innovation capability and entrepreneurial success (indirect effect $\beta = 0.409$, $p < 0.001$), with no significant direct path from AI capability to success when resilience is included in the model. This finding fundamentally challenges the technology-performance determinism prevalent in information management and digital innovation literature, which often presumes direct pathways from technological capabilities to performance outcomes (Nambisan et al., 2017; Piccoli et al., 2024). Prior research on AI and digital transformation has frequently treated technology adoption as a primary driver of competitive advantage, implicitly assuming that advanced information processing capabilities translate directly into superior performance (Gama & Magistretti, 2025; Simsek et al., 2024). Our findings demonstrate that this assumption oversimplifies the complex organizational transformation required to extract value from AI investments.

This result extends dynamic capability theory (Barreto, 2010; Cavusgil & Deligonul, 2024; Teece et al., 1997) by empirically demonstrating the sequential transformation process through which technological information processing potential (AI innovation capability) must first convert into organizational adaptive capacity (strategic resilience) before enabling performance outcomes. Unlike prior dynamic capability research emphasizing parallel development of technological and organizational capabilities (Helfat & Peteraf, 2003), our findings reveal a necessary temporal sequencing where resilience development serves as a prerequisite for AI value capture. This challenges recent conceptualizations of digital transformation that emphasize "digital resource primacy" (Piccoli et al., 2024), suggesting instead that organizational adaptive capacity holds primacy in early-stage ventures, with digital resources serving as enablers rather than primary drivers. For resource-constrained startups, this implies that AI investments without concurrent resilience-building represent misallocated resources—a counterintuitive insight given practitioner emphasis on rapid technology adoption (Browder et al., 2024; Chattopadhyay et al., 2024). The mediation mechanism also contradicts earlier information management scholarship focusing on digital innovation in established firms, which emphasized the role of pre-existing information resources but devoted limited attention to how younger ventures must first build adaptive capacity before leveraging intelligent information systems (Nambisan et al., 2017; Pedota, 2023).

Having established that resilience serves as a necessary mediating pathway for AI value capture (Finding 1), we now turn to a critical performance gap revealed by IPMA: despite its demonstrated mediating primacy, strategic resilience is systematically underdeveloped relative to AI capability in our sample—a misalignment with significant strategic and resource-allocation implications (Finding 2).

8.2. Finding 2: importance-performance gap—resilience as undervalued strategic priority

Our IPMA results reveal a critical performance gap: strategic resilience demonstrates substantially higher importance for entrepreneurial success (total effects = 0.836) compared to AI innovation capability (total effects = 0.561), yet exhibits lower performance levels (58.934 vs. 60.987), indicating that ventures have prioritized technological capability development over resilience capacity building. This finding contradicts the prevailing entrepreneurship literature's emphasis on technological innovation and digital capability development as primary strategic imperatives for early-stage ventures (Ameen et al., 2024; Gama & Magistretti, 2025; Simsek et al., 2024). The technology-centric narrative dominating digital entrepreneurship research implicitly encourages ventures to prioritize AI adoption, integration, and application innovation—capabilities that, according to our findings, deliver comparatively less performance impact than resilience capabilities. This misalignment between scholarly emphasis and empirical importance-performance patterns suggests that current research may inadvertently misdirect entrepreneurial resource allocation decisions.

Moreover, this finding challenges the resource-based view's traditional emphasis on rare, valuable, and inimitable resources—typically technological assets—as primary sources of competitive advantage (Barney, 1991), demonstrating instead that adaptive organizational capabilities, which are process-based, socially complex, and difficult to imitate, provide greater performance leverage in volatile entrepreneurial contexts (Browder et al., 2024; Lengnick-Hall et al., 2011). This importance-performance asymmetry extends organizational resilience theory by providing quantitative evidence that resilience serves as a more critical higher-order performance driver than technological capabilities in resource-constrained settings (Engelen et al., 2024). While resilience literature has theoretically positioned adaptive capacity as fundamental to survival and growth, empirical demonstrations of resilience's performance primacy over technology capabilities remain scarce, particularly in AI-intensive contexts. Our findings suggest a theoretical reorientation: rather than viewing resilience as a defensive capability activated during crises (Sahasranamam and Soundararajan, 2022; Scheidgen et al., 2024), it should be conceptualized as a proactive value-creation mechanism that amplifies returns from technological investments during both normal operations and disruptions. The performance gap further implies systematic underinvestment in resilience development—a strategic blind spot that may explain why many AI initiatives fail to generate expected returns (Ameen et al., 2024). For theory development, this suggests incorporating resilience metrics alongside technology metrics in performance models, challenging single-capability perspectives that dominate current research (Al Halbusi et al., 2025; Mikalef & Gupta, 2021).

While the importance-performance gap (Finding 2) identifies strategic resilience as the underinvested strategic priority at the average level, it does not reveal the specific capability profiles through which individual ventures can most efficiently close this gap given their unique resource endowments and competitive contexts. Our fsQCA analysis addresses this limitation by demonstrating that multiple equifinal configurations—involving different combinations of AI capabilities and resilience dimensions—can all lead to high entrepreneurial success, providing strategic flexibility for resource-constrained ventures (Finding 3).

8.3. Finding 3: equifinality—multiple pathways to success through capability configurations

Our fsQCA analysis identifies six distinct configurations of AI capabilities and resilience elements that lead to high entrepreneurial success (solution coverage = 0.633, consistency = 0.861), demonstrating that ventures can achieve equifinal outcomes through fundamentally different strategic pathways—some emphasizing comprehensive resilience with narrow AI profiles (Configurations 1-2), others combining specific AI capabilities with selective resilience dimensions (Configurations 3-6). This finding challenges the linear, universalistic assumptions underlying much of the technology adoption and entrepreneurship literature, which often seeks to identify single "best practices" or universal success factors applicable across contexts (Davidsson et al., 2023; Nambisan et al., 2017). The emergence of multiple viable pathways contradicts deterministic models suggesting that specific AI capabilities (e.g., application innovation) or resilience dimensions (e.g., cognitive elements) are universally necessary or sufficient for success.

Prior research on AI innovation capabilities has predominantly employed variance-based methods (PLS-SEM, regression) that assume symmetric, linear relationships and uniform effects across organizations

(Ameen et al., 2024; Mikalef & Gupta, 2021), potentially overlooking the configurational complexity our fsQCA reveals. The finding that ventures can succeed with Configuration 4 (balanced AI capability without strong application innovation) directly challenges assumptions in innovation management literature that radical innovation or novel applications are essential for entrepreneurial success in technology-intensive contexts (Cillo & Rubera, 2024; Gama & Magistretti, 2025). Similarly, Configuration 6's demonstration that ventures can succeed without strong contextual resilience (stakeholder relationships, resource access) challenges network-centric entrepreneurship theories emphasizing external relationships as fundamental success determinants (Stam & Elfring, 2008; Venkatesh et al., 2017).

These equifinal pathways extend dynamic capability theory by demonstrating that capability orchestration—the specific combination and sequencing of capabilities—matters as much as capability possession (Cavusgil & Deligonul, 2024; Teece, 2007). It is important to note that all six configurations identified by fsQCA represent sufficient conditions for high entrepreneurial success within our sample—that is, they describe characteristic capability profiles observed among top-performing ventures. These configurations do not imply that the identified substitution and complementarity effects generalize to all ventures, nor do they suggest that any single configuration represents a universal prescription. Rather, they reveal the diverse ways in which high-performing ventures in our sample have strategically combined AI capabilities and resilience dimensions to achieve strong outcomes.

Moreover, these fsQCA findings generate knowledge that is categorically distinct from—and irreducible to—what PLS-SEM and IPMA provide. PLS-SEM and IPMA answer the questions: "On average, does the mediation relationship hold?" and "Which construct is the higher priority for investment?" fsQCA answers a fundamentally different question: "What specific combinations of conditions are jointly sufficient for high success?" This distinction matters because: (a) no single condition is universally necessary for success—each appears absent in at least one high-success configuration, demonstrating that the "importance" identified by IPMA is an average-level pattern that masks substantial configurational heterogeneity; (b) substitution effects invisible to SEM are revealed—for example, Configuration 5 demonstrates that strong contextual resilience can compensate for absent cognitive resilience; (c) equifinality means that a startup with strong integration and application AI capabilities but underdeveloped management capabilities (Configuration 3) requires different strategic advice than one with strong management AI and comprehensive resilience (Configuration 1), even though both achieve high success; and (d) conjunctural causation—conditions that must co-occur—is identified, which additive regression logic systematically underestimates. Furthermore, the fsQCA configurations map directly onto the AI adoption profiles documented in 3.2: information technology ventures deploying AI primarily for product development align with Configurations 3 and 5 (application innovation prominent), while manufacturing ventures using AI for supply chain optimization align with Configurations 1 and 4 (management innovation prominent). This industry-configuration alignment enriches the practical guidance by connecting abstract theoretical configurations to concrete organizational contexts.

While the configurational diversity revealed by fsQCA (Finding 3) establishes that multiple capability profiles can lead to high entrepreneurial success, it does not reveal the organizational processes that actually govern how ventures enact these different configurations. Our qualitative evidence addresses this limitation by revealing that strategic resilience operates not as a passive structural attribute but as an active, dynamic sense-making and adaptation process that determines how ventures translate AI-generated information into competitive action (Finding 4).

8.4. *Finding 4: processual dynamics—qualitative evidence of resilience as active transformation mechanism*

Our semi-structured interviews (Study 4) reveal that strategic resilience operates not as a passive buffer but as an active transformation mechanism through which ventures convert AI-enabled insights into adaptive organizational responses. Interviewees consistently emphasized that AI value emerged only when leaders proactively leveraged AI outputs to accelerate organizational learning, facilitate cross-functional collaboration, and enable continuous strategic refinement—processes that embody resilience's cognitive, behavioral, and contextual dimensions (Table 9). This processual insight challenges the predominant treatment of both AI capabilities and resilience as relatively static organizational attributes or resources in quantitative research (Browder et al., 2024; Lengnick-Hall et al., 2011; Mikalef & Gupta, 2021). Much of the existing literature conceptualizes AI capabilities through structural indicators (systems adopted, integration breadth, management applications) and resilience through outcome measures (survival rates, recovery speed), inadvertently obscuring the dynamic, interactive processes through which these capabilities co-evolve and mutually reinforce each other over time (Davidsson et al., 2023; Williams et al., 2021).

Our interview evidence reveals that resilience is not merely an outcome of AI capability development but rather an ongoing sense-making and adaptation process that shapes how ventures interpret AI-generated information, guiding decisions on which insights warrant action, mobilize resources for implementation, and learn from experimentation outcomes. This process perspective contradicts resource-based views treating capabilities as possessions to be accumulated (Barney, 1991) and aligns more closely with practice-based and evolutionary perspectives emphasizing capabilities as ongoing accomplishments requiring continuous enactment (Williams et al., 2021). The finding that "short-term gains from AI projects require continuous refinement of strategic approaches" (as noted by Interviewees 4 and 10) directly challenges the project-based, milestone-driven approaches to AI implementation prevalent in practitioner literature, which often treat AI deployment as discrete initiatives with defined endpoints rather than continuous adaptation processes (Goto, 2023; Hendriksen, 2023).

This processual finding extends information management theory by revealing how AI-enabled information flows undergo organizational translation through resilience processes before generating performance value (Nambisan et al., 2017; Pedota, 2023). While information management research has long recognized that raw data requires transformation into actionable knowledge, the specific organizational processes mediating this transformation in AI-intensive contexts have remained underspecified. Our interview evidence suggests that resilience provides the interpretive and implementation infrastructure through which AI-generated predictions, patterns, and recommendations become embedded in organizational routines, strategic decisions, and operational adjustments. This perspective challenges technology-deterministic views suggesting AI autonomously improves decision-making (Carter & Liu, 2025) and instead positions AI as a decision-support resource whose value depends on organizational interpretation, contestation, and integration—processes fundamentally shaped by resilience capabilities. The finding also illuminates temporal dynamics often obscured in cross-sectional research: ventures must repeatedly cycle through sensing (AI-enabled pattern detection), seizing (resilience-enabled opportunity identification), and reconfiguring (resilience-enabled resource redeployment) to sustain AI value over time (Teece, 2007). This continuous cycling contradicts static capability accumulation models and suggests that ventures face ongoing reinvestment requirements in both AI and resilience development to prevent capability atrophy—an insight with implications for understanding why initial AI successes frequently fail to compound into sustained advantages (Salvato et al., 2020; Scheidgen et al., 2024).

8.5. Cross-context robustness and institutional nuances

Our study's dual-context design, sampling ventures from both China and Europe, warrants further discussion in light of these four findings. While the statistical control for region did not yield significant differences in the core path model, this finding itself is noteworthy. It suggests that the fundamental mechanism—whereby AI innovation capability enhances entrepreneurial success through the development of strategic resilience—is a robust phenomenon that transcends these distinct institutional settings. However, this does not preclude the possibility of nuanced differences in the enactment of these capabilities. For instance, China's strong state-driven support for AI and its dynamic digital ecosystem might foster more rapid AI adoption and application innovation (Configuration 5's application-driven pathway may be more prevalent). In contrast, Europe's mature regulatory landscape, including stringent data privacy laws, may lead firms to place a greater emphasis on AI management innovation and the contextual elements of resilience, such as stakeholder trust (Configurations 2 and 4 emphasizing integration and management with strong contextual resilience may be more common). These institutional variations suggest that while the core mediation mechanism operates universally, the specific configurations through which ventures achieve success (Finding 3) may exhibit regional clustering patterns. This interpretation is further supported by the AI adoption profiles documented in 3.2: China's state-led AI ecosystem, which facilitates rapid scaling of AI application innovations, is consistent with a higher prevalence of Configuration 5 (Application-Driven Contextual Adaptation) among Chinese ventures, where government-supported stakeholder networks provide the contextual resilience that complements aggressive AI application deployment. Europe's GDPR-constrained environment, by contrast, may favor Configurations 2 and 4, where careful AI integration and management innovation—rather than radical AI application—are combined with strong cognitive and contextual resilience elements that enable compliant yet adaptive AI governance. These technology-context alignments represent a productive direction for future comparative research. Future research could benefit from a more granular comparative analysis, perhaps using qualitative methods, to explore how these different environmental pressures shape the specific manifestation of information-driven resilience and influence which of the six identified configurations prove most viable in each institutional context (Cavusgil and Deligonul, 2024; Zahra et al., 2024).

9. Implications

9.1. Theoretical implications

This study reveals how artificial intelligence innovation capability stimulates novel forms of strategic resilience, extending theoretical perspectives on the co-evolution of information innovation and dynamic knowledge capabilities (Nambisan et al., 2017; Teece et al., 1997). The findings indicate that information-intensive ventures enhance adaptive knowledge processes not merely through technical expertise but by orchestrating intangible information resources that foster rapid knowledge reconfiguration, contributing to a more nuanced understanding of how information management competencies amplify organizational knowledge agility (Lengnick-Hall et al., 2011). This approach enriches current debates on whether rapid information technology adoption alone secures knowledge performance gains, highlighting that the synergistic effect of information-driven insights, knowledge leadership vision, and organizational information resilience underpins sustained knowledge advantages.

These results extend understanding of information capability-building processes by demonstrating how artificial intelligence enables ventures to sense, select, and seize emerging knowledge opportunities under heightened information uncertainty (Ameen et al., 2024; Gama & Magistretti, 2025). The empirical evidence broadens prior conceptions that information transformation primarily benefits large incumbents, showing that early-stage firms also shape new competitive information terrains by aligning information technology initiatives with strategic knowledge resilience routines (Browder et al., 2024). This work indicates that information-based entrepreneurship research should further examine cross-level interactions among founding teams, knowledge ecosystem stakeholders, and advanced information solutions to articulate how adaptive learning unfolds across volatile information contexts.

9.2. Practical implications

Organizations seeking to harness artificial intelligence for information management advantage should implement iterative information experimentation that integrates machine intelligence with frequent knowledge feedback loops, thereby enabling near-real-time adaptation to evolving information demands (Engelen et al., 2024; Teece, 2007). Information managers can prioritize agile knowledge decision protocols that empower cross-functional teams to redesign information workflows as new data emerges, creating a structured yet adaptive process for identifying and resolving information bottlenecks (Gama & Magistretti, 2025). The continuous refinement of information prototype solutions thus ensures rapid alignment between technological information possibilities and knowledge-based validations.

Cross-functional information training activities that familiarize personnel with data analytics, knowledge design thinking, and information project management can translate abstract artificial intelligence insights into knowledge products or process innovations (Ameen et al., 2024; Cillo & Rubera, 2024). This approach reduces information change resistance by enabling shared mental models of how system information outputs should guide knowledge decisions, thereby fostering a more unified organizational response during external information disruptions (Lengnick-Hall et al., 2011). Organizations can further implement small-scale information pilot projects to test emergent knowledge ideas in a low-risk environment, accelerating organizational learning before committing information resources at scale.

9.3. Policy implications

Our dual-context research design spanning China and Europe reveals that effective policy interventions must be tailored to regional institutional contexts and differentiated by stakeholder roles. In China's state-led AI ecosystem, policymakers should leverage centralized coordination infrastructure to create national-level AI innovation capability development programs, establishing capability assessment frameworks, government-sponsored AI resource pools (computing infrastructure, datasets, algorithmic libraries), and staged funding mechanisms that prioritize startups demonstrating strong cognitive and behavioral resilience elements identified in our fsQCA configurations (Zahra et al., 2024). Provincial and municipal governments should establish regional AI resilience centers providing integrated technical AI capability development and strategic resilience coaching, recognizing from our findings that AI capability alone is insufficient without accompanying resilience development (Browder et al., 2024; Simsek et al., 2024). In Europe's GDPR-constrained regulatory landscape, policymakers must balance innovation enablement with compliance requirements through five complementary strategies: developing AI regulatory sandboxes for early-stage ventures with temporary exemptions from certain requirements, creating cross-border capability development programs navigating heterogeneous regulations, establishing GDPR-compliant federated data-sharing consortia, providing regulatory compliance vouchers subsidizing legal expertise, and incentivizing privacy-by-design AI solutions through grants and procurement preferences. Our fsQCA findings showing that contextual resilience elements (stakeholder relationships, resource access) play critical roles in multiple high-success configurations suggest that European policies should particularly emphasize building trust-based stakeholder networks—including data protection authorities, industry consortia, and research institutions—that facilitate compliant AI innovation (Engelen et al., 2024; Nambisan et al., 2017).

Incubators and accelerators should redesign support programs to explicitly address the AI capability-resilience nexus revealed in our mediation results (H3: $\beta = 0.409$, $p < 0.001$), implementing dual-track mentoring programs pairing AI technical expertise with strategic resilience coaching, creating cohort-based learning communities organized around our six identified fsQCA configurations, and developing diagnostic tools based on our three-dimensional frameworks to help startups identify development priorities. Regulatory authorities should adopt differentiated approaches acknowledging the heterogeneity in AI adoption pathways, implementing capability-based regulatory tiers where early-stage ventures with developing capabilities face lighter-touch oversight, creating fast-track approval processes for domains where startups demonstrate both AI capability and resilience indicators, and establishing feedback mechanisms enabling constructive dialogue about regulatory impacts on capability development. The emergence of six distinct configurations leading to high entrepreneurial success (solution coverage = 0.633, consistency = 0.861) demonstrates that ventures achieve success through multiple pathways, suggesting that one-size-fits-all regulatory approaches may inadvertently constrain beneficial diversity and that regulations should scale oversight intensity to organizational capacity while maintaining core safety standards (Browder et al., 2024; Scheidgen et al., 2024).

Investment decision-makers should recalibrate evaluation frameworks based on our finding that AI innovation capability's influence on entrepreneurial success is mediated by strategic resilience, with due diligence assessing not only AI technical capabilities but also cognitive, behavioral, and contextual resilience elements. Our IPMA analysis revealing that strategic resilience demonstrates higher importance (0.836) for entrepreneurial success compared to AI innovation capability (0.561) underscores that resilience assessment should receive commensurate attention in investment decisions. Funding mechanisms should support development timelines allowing ventures to build both AI capabilities and resilience capacity concurrently, incorporate resilience milestones alongside technical milestones in staged investment structures, recognize the equifinality revealed in our fsQCA results by supporting diverse configurations rather than imposing uniform templates, and provide value-added support helping portfolio companies develop underdeveloped dimensions. Investors can facilitate targeted capability development by connecting portfolio companies with technical experts for AI capability dimensions, executive coaches for cognitive resilience, operational consultants for behavioral resilience, and network brokers for contextual resilience, thereby actively contributing to the capability-resilience development process rather than passively evaluating it (Gama & Magistretti, 2025). These integrated policy interventions across regional contexts and stakeholder roles can strengthen information innovation ecosystems where smaller firms, despite resource constraints, access strategic information inputs for building adaptive knowledge capabilities that translate into sustained entrepreneurial success (Ameen et al., 2024; Barreto, 2010).

10. Conclusion, limitations, and outlook

This study advances understanding of how artificial intelligence innovation capability and strategic resilience jointly drive entrepreneurial success in uncertain information contexts. The findings illustrate that integrating artificial intelligence into core information processes is not merely a technological pursuit but rather a means of building adaptive information capacity, facilitating rapid knowledge reconfiguration and proactive information opportunity recognition. Strategic resilience emerges as a central conduit for transforming technological information investments into sustained knowledge growth, thereby underscoring the importance of information-based dynamic capability development in early-stage ventures. These insights contribute to broader discussions on how novel information management competencies can enable firms to navigate information turbulence effectively and remain vigilant to ongoing knowledge shifts, thus extending prior scholarship on both information-driven entrepreneurship and organizational knowledge adaptation.

Several boundary conditions qualify these findings. First, our purposive sampling of ventures already implementing AI introduces selection and survivorship constraints: the relationships we identify characterize the AI-adopting population and may not generalize to ventures that never adopted AI or abandoned implementation (Browder et al., 2024). Second, although our two-wave design temporally separates predictors from outcomes, both AI innovation capability and strategic resilience were collected from the same respondents at Wave 1, meaning the H1 path coefficient may carry residual common method inflation despite satisfactory ex-post controls (Podsakoff et al., 2003). Third, the one-year entrepreneurial success window captures short-to-medium term growth outcomes—revenue, market share, and customer base expansion—rather than longer-term profitability or survival, representing a deliberate scope condition (Davidsson et al., 2023; Richard et al., 2009). Fourth, the formative dimension weights reported for AI innovation capability and strategic resilience reflect sample-specific relative contributions that may shift across more mature ventures, alternative industries, or institutionally homogeneous settings (Hair et al., 2024; Mikalef & Gupta, 2021). Fifth, the non-significant binary regional control (China vs. Europe) should be interpreted cautiously, as a dichotomous indicator compresses substantial within-region institutional heterogeneity and may lack the granularity to detect subtler moderating effects (Cavusgil & Deligonul, 2024; Zahra et al., 2024). Finally, the model's generalizability is intentionally bounded to early-stage, AI-enabled ventures; later-stage firms with greater organizational inertia, legacy systems, and abundant resources would likely exhibit different relationship magnitudes.

Future research should address these limitations along several avenues. Multi-wave panel designs that re-measure AI innovation capability and strategic resilience over time would enable dynamic examination of capability co-evolution and strengthen causal inference (Davidsson et al., 2023; Williams et al., 2021). Multi-informant designs—where technical personnel assess AI capability and strategic managers assess resilience—would provide stronger CMV-protected estimates. Extended longitudinal studies tracking three-to-five year outcomes would clarify whether strategic resilience continues to mediate AI's performance impact as ventures mature and transition from early-stage to growth-stage operations. Researchers could also refine entrepreneurial success measurement by incorporating fine-grained longitudinal indicators capturing knowledge performance volatility in rapidly evolving information ecosystems and exploring cross-level factors such as inter-firm knowledge collaborations or sociocultural influences that moderate the AI capability–resilience interplay. Additionally, multi-country comparative case studies examining how institutional, regulatory, and cultural factors—including contrasting frameworks such as GDPR versus China's data governance regimes, state-led versus market-driven innovation ecosystems, and cultural dimensions such as uncertainty avoidance and long-term orientation—shape the specific practices through which ventures develop AI capabilities and strategic resilience would extend the boundary conditions of our model (Cavusgil & Deligonul, 2024; Zahra et al., 2024).

Data availability statement

The datasets used or analyzed during the current study are available from the first author on reasonable request. The semi-structured interview protocol is provided in the Supporting Materials.

CRediT authorship contribution statement

Hao Zhang: Writing – review & editing, Writing – original draft.

Shaofeng Wang: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization.

Ethics approval

Not applicable.

Funding

This work was supported by National Social Science Fund of China Project [21&ZD322].

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled "*Artificial Intelligence for Information-Driven Resilience: Enhancing Strategic Adaptation and Entrepreneurial Success*".

Acknowledgment

Thank all respondents for participating in this survey. Thanks to our colleagues and Digital Tools for their help with the article's language.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijinfomgt.2026.103057.

Data availability

Data will be made available on request.

References

Al Halbusi, H., Al-Sulaiti, K. I., Alalwan, A. A., & Al-Busaidi, A. S. (2025). AI capability and green innovation impact on sustainable performance: Moderating role of big data and knowledge management. *Technological Forecasting and Social Change, 210*, Article 123897.
Ameen, N., Tarba, S., Cheah, J. H., Xia, S., & Sharma, G. D. (2024). Coupling artificial intelligence capability and strategic agility for enhanced product and service creativity. *British Journal of Management, 35*(4), 1916–1934.
Armstrong, J. S., & Overton, T. S. (1977). Estimating nonresponse bias in mail surveys. *Journal of Marketing Research, 14*(3), 396–402.
Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management, 17*(1), 99–120.
Barreto, I. (2010). Dynamic capabilities: A review of past research and an agenda for the future. *Journal of Management, 36*(1), 256–280.
Bawack, R., Kala Kamdjoug, J. R., Agnia Nkolo, P. M., Bonhoure, E., & Ouallet, S. (2025). Strategic CSR: investigating the ripple effect of corporate social responsibility on perceived marketing performance through corporate image and reputation. *Journal of Strategic Marketing, 33*(5), 639–657. https://doi.org/10.1080/0965254X.2025.2508706
Breidbach, C. F., Joshi, A. M., Twigg, A., & Dickens, G. (2024). Orchestrating Digital Resilience: A Clinical IS Study of an Everything-as-a-Service Technology Strategy. *European Journal of Information Systems*, 1–18.
Browder, R. E., Dwyer, S. M., & Koch, H. (2024). Upgrading adaptation: How digital transformation promotes organizational resilience. *Strategic Entrepreneurship Journal, 18*(1), 128–164.
Carter, L., & Liu, D. (2025). How was my performance? Exploring the role of anchoring bias in AI-assisted decision making. *International Journal of Information Management, 82*, Article 102875.
Cavusgil, S. T., & Deligonul, S. Z. (2024). Dynamic capabilities framework and its transformative contributions. *Journal of International Business Studies*, 1–10.
Chattopadhyay, S., Honoré, F., & Won, S. (2024). Free range startups? Market scope, academic founders, and the role of general knowledge in AI. *Strategic Management Journal*. https://doi.org/10.1002/smj.3685
Chau, M., & Xu, J. (2025). An IS research agenda on large language models: development, applications, and impacts on business and management. *ACM Transactions on Management Information Systems, 16*(1), 1–11.
Cillo, P., & Rubera, G. (2024). Generative AI in innovation and marketing processes: A roadmap of research opportunities. *Journal of the Academy of Marketing Science*, 1–18.
Davidsson, P., Recker, J., Chalmers, D., & Carter, S. (2023). Environmental change, strategic entrepreneurial action, and success: Introduction to a special issue on an important, neglected topic. *Strategic Entrepreneurship Journal, 17*(2), 322–334.
Diamantopoulos, A., & Siguaw, J. A. (2006). Formative versus reflective indicators in organizational measure development: A comparison and empirical illustration. *British Journal of Management, 17*(4), 263–282.
Diamantopoulos, A., & Winklhofer, H. M. (2001). Index construction with formative indicators: An alternative to scale development. *Journal of Marketing Research, 38*(2), 269–277.
Engelen, A., Huesker, C., Rieger, V., & Berg, V. (2024). Building a resilient organization through a pre-shock strategic emphasis on innovation. *Journal of Product Innovation Management, 41*(1), 36–61.
Fiss, P. C. (2011). Building better causal theories: A fuzzy set approach to typologies in organization research. *Academy of Management Journal, 54*(2), 393–420.
Galindo-Martin, M. A., Mendez-Picazo, M. T., & Perez-Pujol, R. S. (2025). Open innovation and sustainable development: A micro and macroeconomic analysis using a mixed method research with PLS-SEM-NCA and Delphi. *International Journal of Information Management, 82*, Article 102874.
Gama, F., & Magistretti, S. (2025). Artificial intelligence in innovation management: A review of innovation capabilities and a taxonomy of AI applications. *Journal of Product Innovation Management, 42*(1), 76–111.
Goto, M. (2023). Anticipatory innovation of professional services: The case of auditing and artificial intelligence. *Research Policy, 52*(8), Article 104828.
Gupta, S., Wang, Y., Patel, P., & Czinkota, M. (2025). Navigating the future of AI in marketing: AI integration across borders, ethical considerations, and policy implications. *International Journal of Information Management, 82*, Article 102871.
Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2022). *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)* (3rd ed.). Thousand Oaks, CA: Sage.
Hair, J. F., Sarstedt, M., Ringle, C. M., & Gudergan, S. P. (2024). *Advanced Issues in Partial Least Squares Structural Equation Modeling (PLS-SEM)* (2nd ed.). Thousand Oaks, CA: Sage.
Helfat, C. E., & Peteraf, M. A. (2003). The dynamic resource-based view: Capability lifecycles. *Strategic Management Journal, 24*(10), 997–1010.
Hendriksen, C. (2023). Artificial intelligence for supply chain management: Disruptive innovation or innovative disruption? *Journal of Supply Chain Management, 59*(3), 65–76.
Hora, M., & Dutta, D. K. (2013). Entrepreneurial firms and downstream alliance partnerships: Impact of portfolio depth and scope on technology innovation and commercialization success. *Production and Operations Management, 22*(6), 1389–1400.
Hu, D., Lan, Y., Yan, H., & Chen, C. W. (2025). What makes you attached to social companion AI? A two-stage exploratory mixed-method study. *International Journal of Information Management, 83*, Article 102890.
Jarvis, C. B., MacKenzie, S. B., & Podsakoff, P. M. (2003). A critical review of construct indicators and measurement model misspecification in marketing and consumer research. *Journal of Consumer Research, 30*(2), 199–218.
Jaskiewicz, P., Combs, J. G., & Rau, S. B. (2015). Entrepreneurial legacy: Toward a theory of how some family firms nurture transgenerational entrepreneurship. *Journal of Business Venturing, 30*(1), 29–49.
Ku, E. C., & Chen, C. D. (2024). Artificial intelligence innovation of tourism businesses: From satisfied tourists to continued service usage intention. *International Journal of Information Management, 76*, Article 102757.
Lengnick-Hall, C. A., Beck, T. E., & Lengnick-Hall, M. L. (2011). Developing a capacity for organizational resilience through strategic human resource management. *Human Resource Management Review, 21*(3), 243–255.
Li, Y., Wang, X., Gong, T., & Wang, H. (2023). Breaking out of the pandemic: How can firms match internal competence with external resources to shape operational resilience? *Journal of Operations Management, 69*(3), 384–403.
Li, N., Yao, Q., Tang, H., & Lu, Y. (2025). Is digitalization necessary? Configuration of supply chain capabilities for improving enterprise competitive performance. *Journal of Business Research, 186*, Article 114972.
Ma, L., Yu, P., Zhang, X., Wang, G., & Hao, F. (2024). How AI use in organizations contributes to employee competitive advantage: The moderating role of perceived organization support. *Technological Forecasting & Social Change, 209*(12), Article 123801.
Mikalef, P., & Gupta, M. (2021). Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance. *Information & Management, 58*(3), Article 103434.
Nambisan, S., Lyytinen, K., Majchrzak, A., & Song, M. (2017). Digital innovation management: Reinventing innovation management research in a digital world. *MIS Quarterly, 41*(1), 223–238.
Pedota, M. (2023). Big data and dynamic capabilities in the digital revolution: The hidden role of source variety. *Research Policy, 52*(7), Article 104812.
Peretz-Andersson, E., Tabares, S., Mikalef, P., & Parida, V. (2024). Artificial intelligence implementation in manufacturing SMEs: A resource orchestration approach. *International Journal of Information Management, 77*, Article 102781.
Piccoli, G., Grover, V., & Rodriguez, J. (2024). Digital transformation requires digital resource primacy: Clarification and future research directions. *The Journal of Strategic Information Systems, 33*(2), Article 101835.
Pinky, A. S., Talukder, T., Chowdhury, S. N., Hera, A., Khan, N. A., Faruque, M. O., & Ali, M. J. (2024). Information Technology Acceptance: Construct Development and Empirical Validation. *American Journal of Information Systems, 9*(1), 11–18.

Podsakoff, P. M., MacKenzie, S. B., Lee, J., & Podsakoff, N. P. (2003). Common method biases in behavioral research. *Journal of Applied Psychology, 88*(5), 879–903.

Przegalinska, A., Triantoro, T., Kovbasiuk, A., Ciechanowski, L., Freeman, R. B., & Sowa, K. (2025). Collaborative AI in the workplace: Enhancing organizational performance through resource-based and task-technology fit perspectives. *International Journal of Information Management, 81*, Article 102853.

Ragin, C. C. (2008). *Redesigning social inquiry: Fuzzy sets and beyond*. University of Chicago Press.

Ragin, C. C., & Davey, S. (2023). *fsQCA (Version 4.1) [Computer software]*. Irvine, CA: University of California.

Ren, Y., Li, J. J., Zhao, W., & Zhang, L. (2024). The Delicate Equilibrium: Unveiling the Curvilinear Nexus Between Supply Concentration and Organizational Resilience. *British Journal of Management*. https://doi.org/10.1111/1467-8551.12800

Richard, P. J., Devinney, T. M., Yip, G. S., & Johnson, G. (2009). Measuring organizational performance: Towards methodological best practice. *Journal of Management, 35*(3), 718–804.

Roppelt, J. S., Schuster, A., Greimel, N. S., Kanbach, D. K., & Sen, K. (2025). Towards effective adoption of artificial intelligence in talent acquisition: A mixed method study. *International Journal of Information Management, 82*, Article 102870.

Sahasranamam, S., & Soundararajan, V. (2022). Innovation ecosystems: what makes them responsive during emergencies? *British Journal of Management, 33*(1), 369–389.

Salvato, C., Sargiacomo, M., Amore, M. D., & Minichilli, A. (2020). Natural disasters as a source of entrepreneurial opportunity: Family business resilience after an earthquake. *Strategic Entrepreneurship Journal, 14*(4), 594–615.

Scheidgen, K., Günzel-Jensen, F., & Schmidt, S. L. (2024). Entrepreneurial Resourcefulness Throughout Crisis. *Entrepreneurship Theory and Practice*. https://doi.org/10.1177/10422587241269067

Scholz, R. W., Czichos, R., Parycek, P., & Lampoltshammer, T. J. (2020). Organizational vulnerability of digital threats: A first validation of an assessment method. *European Journal of Operational Research, 282*(2), 627–643.

Seyfi, S., Kimbu, A. N., Tavangar, M., Vo-Thanh, T., & Zaman, M. (2025). Surviving crisis: Building tourism entrepreneurial resilience as a woman in a sanctions-ravaged destination. *Tourism Management, 106*, Article 105025.

Simsek, Z., Heavey, C., König, A., & Stam, W. (2024). Leading digital transformation in incumbent firms: A strategic entrepreneurship framing. Strategic Entrepreneurship. *Journal, 18*(1), 91–102.

Stam, W., & Elfring, T. (2008). Entrepreneurial orientation and new venture performance: The mediating role of network strategies. *Academy of Management Journal, 51*(1), 97–111.

Staniewski, M., Awruk, K., Leonardi, G., & Słomski, W. (2025). Family communication and entrepreneurial success–The mediating role of entrepreneurial self-efficacy. *Journal of Innovation & Knowledge, 10*(1), Article 100635.

Strauss, L. M., Klein, A. Z., & Scornavacca, E. (2024). Adopting emerging information technology: a new affordances process framework. *International Journal of Information Management, 76*, Article 102772.

Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal, 28*(13), 1319–1350.

Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal, 18*(7), 509–533.

Toorajipour, R., Oghazi, P., & Palmié, M. (2024). Data ecosystem business models: Value propositions and value capture with Artificial Intelligence of Things. *International Journal of Information Management, 78*, Article 102804.

Venkatesh, V., Shaw, J. D., Sykes, T. A., Wamba, S. F., & Macharia, M. (2017). Networks, technology, and entrepreneurship: A field quasi-experiment among women in rural India. *Academy of Management Journal, 60*(5), 1709–1740.

Williams, T. A., Zhao, E. Y., Sonenshein, S., Ucbasaran, D., & George, G. (2021). Breaking boundaries to creatively generate value: The role of resourcefulness in entrepreneurship. *Journal of Business Venturing, 36*(5), Article 106141.

Xie, X., & Wu, Y. (2022). Doing well and doing good: How responsible entrepreneurship shapes female entrepreneurial success. *Journal of Business Ethics, 178*(3), 803–828.

Zahra, S. A., Criaco, G., Petricevic, O., & Hashai, N. (2024). Conceptualizing international new ventures as the nexus of entrepreneurship and international business. *Journal of International Business Studies, 55*(8), 1048–1056.